

Laboratory Assessment of Kidney Disease: Glomerular Filtration Rate, Urinalysis, and Proteinuria

Anoushka Krishnan | Adeera Levin

CHAPTER OUTLINE

GLOMERULAR FILTRATION RATE, 733
URINALYSIS, 742
PROTEINURIA, 744

URINE MICROSCOPY, 752
SUMMARY, 756

KEY POINTS

- The glomerular filtration rate (GFR) is the amount of fluid filtered into the Bowman space per unit of time and is dependent on the porosity and the surface area of the glomerular membrane as well as the hydraulic pressure and oncotic pressure on the capillary side and in the Bowman space. Normal GFR is $> 90 \text{ mL/min/1.73 m}^2$. There is a gradual decline in GFR with age, but the effect of age is variable.
- GFR measurement is cumbersome, as it needs the measurement of clearance of an exogenous filtration marker such as inulin. Hence, estimated GFRs (eGFRs) are calculated using the values of plasma creatinine. The most commonly used methods include the Cockcroft–Gault (CG) equation, the Modification of Diet in Renal Disease (MDRD) study equation, and more recently the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. The CKD-EPI equation is more accurate than the MDRD equation and both are more accurate than the CG equation.
- Special formulae may need to be used for children and the elderly, but the full age spectrum method of eGFR calculation may help circumnavigate this problem. Other unique groups include pregnant women, renal transplant recipients, and those with unusual body habitus or muscle mass.
- The main use of eGFR equations is in the detection, monitoring, and prognostication of CKD, which is largely asymptomatic. These equations have been derived from populations with stable CKD and are therefore not useful in the acute setting.
- Creatinine is a widely used marker of renal function. Many factors affect creatinine levels, such as muscle mass/injury, diet, age, gender, physical activity, and race. Creatinine is insensitive to even significant declines in GFR at the upper limit of normal due to the nonlinear relationship between creatinine and GFR. This is due to compensatory hyperfiltration of remaining functioning nephrons.
- Creatinine clearance (CrCl) is measured by collecting the patient's urine for 24 hours and measuring the total amount of excreted creatinine and the volume of urine. CrCl usually tends to overestimate eGFR.
- Proteinuria is one of the earliest markers of renal disease, occurring before a reduction in eGFR is noticed. It imparts diagnostic and prognostic information and is now a part of the Kidney Disease Improving Global Outcomes criteria for stratification of CKD.
- Casts are cylindrical bodies of renal origin that form from the aggregation of fibrils of Tamm–Horsfall glycoprotein (uromodulin). Red cell casts are always pathologic and indicate significant glomerular pathology.

An appreciation of the components of laboratory assessment of kidney disease is essential to the practicing clinician. This chapter describes three key aspects of laboratory assessment: the glomerular filtration rate (GFR), urinalysis, and proteinuria. Issues related to measurement tools, precision, bias, and interpretation are addressed for each of these variables, so that the reader may better appreciate the role of the laboratory in the diagnosis, follow-up, and management of kidney disease. Understanding the physiology of kidney function is essential to the interpretation of laboratory measurements and is highlighted within each section.

GFR is the single best measure of kidney function. Large studies consistently demonstrate the relationship of GFR to outcomes in general and renal populations, and the Kidney Disease Improving Global Outcomes (KDIGO) classification system uses GFR as one of the key dimensions in the diagnosis of chronic kidney disease (CKD). Urinary abnormalities may indicate acute or chronic conditions and isolated kidney or systemic diseases, and can be used to monitor both kidney and systemic diseases. Proteinuria is similarly an important marker of kidney disease and can also be seen in acute or chronic kidney conditions. In patients with CKD, the magnitude of albuminuria is proposed as a second dimension for the classification of severity in the KDIGO classification system because it affects prognosis.

The laboratory assessment of kidney disease can be used to diagnose, prognosticate, and measure progression of disease or response to therapy. CKD is increasingly recognized as an important public health problem, so accurate and appropriate use of laboratory testing is important. This chapter facilitates understanding and interpretation of key tests used in the assessment of kidney disease, both acute and chronic.

GLOMERULAR FILTRATION RATE

“Glomerular filtration rate” describes one of the key roles of the kidney: to filter plasma so as to excrete waste products and produce urine (an ultrafiltrate of plasma). In clinical practice, estimates of GFR are obtained using equations, and direct measurement is reserved for specific circumstances. This section reviews normal physiology, use of various filtration markers, the development and use of equations, and special circumstances in which GFR interpretation needs to be contextualized.

NORMAL PHYSIOLOGY

Separation of an ultrafiltrate of plasma across the barrier composed of the capillary wall, glomerular basement membrane, and podocyte is the first step in the production of urine and multiple functions of the kidney. The amount of fluid filtered into the Bowman space per unit of time is the GFR. Fluid movement is governed by Starling forces and so the glomerular filtrate that is produced is dependent on the following determinants:

- Porosity of the membrane (p)
- Surface area of the membrane (K)
- Hydraulic pressure and oncotic pressure on the capillary side (P_{GC} and π_{GC})

Table 23.1 Factors That May Alter Determinants of Glomerular Filtration Rate

p	Generally less important than other factors
K	↑ by relaxation of mesangial cells ↓ in glomerulonephritis and glomerulosclerosis
P_{GC}	↑ by afferent arteriole dilatation and efferent arteriole constriction
P_{BS}	↑ by raised intratubular pressure due to obstruction
π_{GC}	↑ by raised systemic oncotic pressure or decreased renal plasma flow
π_{BS}	Minimal impact

K , Surface area of the membrane; p , porosity of the glomerular membrane; P_{BS} , hydraulic pressure in the Bowman space; P_{GC} , glomerular capillary hydraulic pressure; π_{BS} , oncotic pressure in the Bowman space; π_{GC} , glomerular capillary oncotic pressure.

- Hydraulic pressure and oncotic pressure in the Bowman space (P_{BS} and π_{BS})

GFR can be calculated using the following equation:

$$GFR = p \times [(P_{GC} - P_{BS}) - K(\pi_{GC} - \pi_{BS})]$$

Factors that may alter these determinants of GFR are shown in Table 23.1. For further discussion of the physiology of glomerular ultrafiltration, please see Chapter 3.

The kidney filters approximately 180 L of plasma per day, which is equivalent to 125 mL/min. Kidney function is proportional to kidney size, which in turn is proportional to body surface area (BSA). Normal GFR is > 90 mL/min/1.73 m². In young adults GFR is approximately 120 to 130 mL/min/1.73 m². There is a gradual decline in GFR with age, but the effect of age is variable, in that some elderly patients have no change in renal function (see Chapter 22).

Protein intake and hyperglycemia can raise GFR by increasing renal plasma flow; the mechanism by which this occurs is unclear but may be activation of the intrarenal renin-angiotensin system.

MEASUREMENT OF GFR

GFR can be measured provided that the concentration of a substance that is freely filtered at the glomerulus and neither reabsorbed nor secreted in the renal tubule is measured in the plasma and in a timed urine collection. GFR is equivalent to the “clearance rate” of this substance or filtration marker, which can be calculated as follows:

$$GFR = \frac{\text{Urine concentration of marker} \times \text{urine volume}}{\text{Plasma concentration of marker} \times \text{duration of urine collection}}$$

Filtration markers may be endogenous (e.g., creatinine, cystatin C) or exogenous (e.g., inulin, iothexol). GFR can be expressed as normalized to BSA to reflect physiologic matching of GFR to kidney size and in turn to BSA (mL/min/1.73 m²), or as an absolute value (mL/min). An absolute value of GFR is useful in the setting of drug dosing, which

is discussed further later. As an alternative to direct measurement of the clearance of a filtration marker, GFR can be estimated using an equation based on the plasma concentration of a filtration marker in steady state.

The most appropriate method for measurement of GFR depends on the purpose for which kidney function is being monitored. Direct measurement of GFR is time-consuming and may require medical supervision or the patient's presence at the hospital, so it is not usually performed as part of everyday practice. An exact value of the GFR is not required for most clinical settings. Understanding the trend within an individual patient's renal function is often what is needed. As a result, plasma urea and creatinine and estimated GFR (eGFR) calculations based on plasma creatinine are most often used. Alternatively, accuracy in the measurement of GFR is important for drug dosing of medications with narrow therapeutic windows (e.g., chemotherapy drugs) and is vital for appropriate selection and approval of live donors for kidney transplantation.

GLOMERULAR FILTRATION MARKERS

The ideal glomerular filtration marker probably does not exist. Such a marker would have the following characteristics:

- Distributed freely and instantaneously throughout the extracellular space
- Not bound to plasma proteins
- Freely filtered at the glomerulus
- Not secreted or reabsorbed at the tubules
- Eliminated wholly by the kidney
- Resistant to degradation
- Easy and inexpensive to measure

An understanding of the limitations of each filtration marker is important to be able to interpret results. Given that kidney disease is often asymptomatic and dependent on the accuracy of laboratory tests, it is imperative that the clinician knows the advantages and disadvantages of each specific test before making clinical decisions.

ENDOGENOUS GLOMERULAR FILTRATION MARKERS

Urea

Urea is not an accurate filtration marker because it is subject to a number of influences in addition to glomerular filtration. The liver produces urea in the urea cycle as a waste product of the digestion of protein. Therefore increased plasma levels may be due to factors independent of renal function, such as increased production in high protein intake, gastrointestinal bleeding due to absorption of amino acids from blood in the gastrointestinal tract, and high catabolic states such as those associated with glucocorticoid therapy.

Low urea levels are seen in decreased protein intake and chronic liver disease due to reduced synthesis of urea.

Urea is readily reabsorbed in the proximal tubule, particularly at low urinary flow rates. As a filtration marker, urea therefore has limited use because, although it is freely filtered, significant reabsorption means that the amount that is excreted is not what was filtered. High levels of urea may not necessarily indicate poor renal function but could instead reflect hypovolemia and renal hypoperfusion.

Serum Creatinine

Creatinine is a product of normal muscle metabolism. Phosphocreatine is a source for replenishment of phosphate when adenosine triphosphate is used by muscle cells. Creatine and phosphocreatine are nonenzymatically converted at an almost steady rate (approximately 2% of total creatine per day) to creatinine. Creatinine is not bound to plasma proteins, being freely filtered by the kidney, but it is unfortunately secreted by the tubules, making it an imperfect filtration marker. Despite its many limitations, however, its measurement is still the test most widely used by physicians to gauge renal function.

Serum creatinine level does not measure GFR but varies inversely with GFR and so is an indirect marker of GFR. The serum creatinine level can be used in estimation equations for GFR, and urinary creatinine clearance (CrCl) can be used to approximate GFR.

Factors affecting creatinine levels unrelated to renal function include muscle mass and/or injury and consumption of meat or creatine. Factors affecting muscle mass, such as age, sex, race, and physical activity, can in turn affect creatinine levels. Women, for example, have lower creatinine levels than men at the same level of GFR. Rhabdomyolysis has also been suggested to cause a greater rise in serum creatinine than other causes of acute kidney injury (AKI), because of the release of preformed creatine and phosphocreatine that is converted into creatinine.

Secretion of creatinine by the renal tubule is affected by

Drugs (decrease secretion): For example, trimethoprim, cimetidine, pyrimethamine, dapsone.

Decreased renal function: Each tubule excretes a higher proportion of creatinine as renal function declines. This secretory process is saturated when the serum creatinine level exceeds 1.5 to 2 mg/dL (132–176 $\mu\text{mol/L}$).

Extrarenal degradation of creatinine in the gut also rises as renal function declines due to bacterial overgrowth in the intestines leading to increased bacterial creatininase activity.¹ This extrarenal clearance may be as much as two-thirds of total daily creatinine excretion.

Sources of Error in Measurement of Creatinine. Clinicians may be unaware of the intricacies and complexities of creatinine measurement and how they may be relevant to day-to-day practice.

Laboratory methods to measure serum creatinine include

- Alkaline picrate (Jaffe method)
- Enzymatic methods
- Isotope dilution mass spectrometry (IDMS)
- High-performance liquid chromatography (HPLC)

There is wide variation in measured creatinine concentration, depending on the laboratory method and the instruments used. In 2003, the College of American Pathologists conducted a survey of 5624 participating laboratories that showed a bias from the reference value of between –0.06 and 0.31 mg/dL or –7% and 34%.² This bias was thought to be predominantly due to differences in instrument calibration among manufacturers. As a result, a creatinine standard reference material was prepared by the National Institute of Standards and

Technology, which is currently being used by almost all major manufacturers for calibration. The method of analysis (Jaffe or enzymatic) should also have minimal bias in comparison with the IDMS reference methodology.

Measurement of creatinine obviously affects measured CrCl and calculation of estimated CrCl or GFR, and therefore it has been recommended as part of the KDIGO initiative guidelines that all creatinine results be traceable to reference materials and methods listed on the Joint Committee for Traceability in Laboratory Medicine database.

The Jaffe method measures creatinine by complexing creatinine with alkaline picrate, followed by measurement with a colorimetric technique. This colorimetric assay may falsely measure normal plasma constituents such as glucose and plasma proteins as creatinine. As a result, the measured creatinine result may be falsely high. The Jaffe method may also report falsely low creatinine levels if there are very high serum bilirubin levels. Modified Jaffe methods attempt to take this into account by removing these interfering chromogens before analysis. This inference is sometimes corrected for by some manufacturers by deducting an estimated value based on average bias from measured results. Currently, the techniques used in most laboratories are modified alkaline picrate and enzymatic methods, but it is recommended that the enzymatic method be adopted because this is more specific.

Although serum creatinine concentration is widely used as a marker of renal function, it is insensitive to even significant declines in GFR at the upper limit of normal due to the nonlinear relationship between creatinine and GFR (Fig. 23.1). This is due to compensatory hyperfiltration of remaining functioning nephrons, secretion of creatinine, and extrarenal elimination of creatinine as the GFR declines. It is consequently a poor screening tool for early kidney disease.

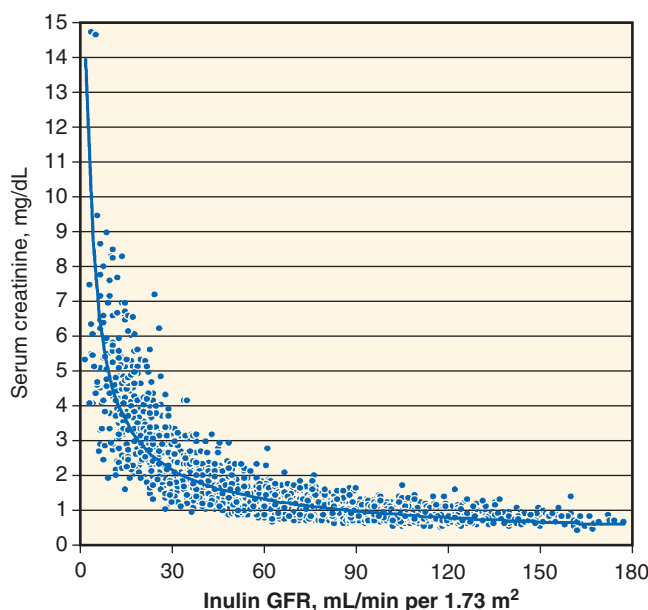


Fig. 23.1 Relationship between plasma creatinine and glomerular filtration rate (GFR). (From Botev R, Mallie JP, Couchoud C, et al. Estimating glomerular filtration rate: Cockcroft-Gault and Modification of Diet in Renal Disease formulas compared to renal inulin clearance. *Clin J Am Soc.* 2009;4:899–906.)

Within-person variability of creatinine is also significant, at 8%; therefore significant change in serum creatinine is generally defined as at least 10%, which at early stages of kidney disease may represent a significant decline in GFR.

Serum creatinine results should be interpreted in the clinical context. The same serum creatinine concentration may correspond to a vastly different GFR in patients of different body composition. Volume status should also be taken into account because dilution of creatinine leads to an apparently low result.

Creatinine Clearance. CrCl is measured by collecting the patient's urine for 24 hours and measuring the total amount of excreted creatinine and the volume of urine. With the equation discussed earlier to calculate clearance of a filtration marker, CrCl is calculated as follows:

For example, for a patient with a serum creatinine value of 100 $\mu\text{mol/L}$, a urine creatinine value of 10,000 $\mu\text{mol/L}$, and a urine volume of 1.44 L, CrCl would be calculated as follows:

$$\begin{aligned}\text{CrCl} &= \frac{\text{urine creatinine concentration} \times \text{urine volume}}{\text{plasma creatinine concentration} \times \text{duration of urine collection}} \\ &= 144 \text{ L}/24 \text{ h} = 100 \text{ mL/min}\end{aligned}$$

If this result is adjusted for BSA in a small person (height = 160 cm, weight = 50 kg, BSA 1.5 m), then the CrCl would be

$$\frac{\text{CrCl} \times 1.73}{\text{BSA}} = \frac{100 \times 1.73}{1.5} = 115 \text{ mL/min}/1.73 \text{ m}^2$$

For a larger person with a BSA of 2.0 m (height = 180 cm, weight = 80 kg), the adjusted CrCl would be 86.5 mL/min/1.73 m².

Although the clearance of an ideal substance would be equivalent to GFR, CrCl tends to exceed true GFR by approximately 10% to 20% because of the secretion of creatinine by the tubules. This error was previously compensated for by errors in the Jaffe assay that would overestimate serum creatinine value. Now that creatinine measurement has been standardized CrCl will consistently overestimate true GFR.

The main problem with CrCl is its reliance on timed urine collection, which is often inaccurate. Furthermore, tubular secretion of creatinine increases with decreased renal function, thus masking a true drop in GFR. In addition, the practical difficulties of conducting a 24-h urine collection for patients and the challenges of handling large volumes of urine for laboratories have resulted in this method being used only infrequently.

Cystatin C

Cystatin C is a low-molecular-weight (LMW; 13-kD) basic protein that is produced at a constant rate by all nucleated cells. It is freely filtered by the glomerulus and is not secreted; proximal tubule cells reabsorb and catabolize the filtered cystatin C so that little is normally excreted in the urine. Therefore although plasma cystatin C levels are used in estimating GFR, cystatin C measurement cannot be used as a conventional urinary excretory marker for GFR. Rather, cystatin C has been regarded as one of several available novel markers of kidney injury.

Plasma cystatin C levels are highest in the first days of life and stabilize after 1 year of age, with levels approximating those of adults. Polymorphic variants in the *CST3* gene encoding cystatin C appear to affect production, and inter-individual variations in cystatin C values account for 25% of its biologic variability, in comparison with 93% for creatinine. Within-person variation of cystatin C values is 6.8%.³

Cystatin C levels were reported to be independent of sex, muscle mass, and age after 12 months, but there is a growing body of evidence that this may not be the case. Cystatin C levels may be affected by factors independent of renal function, such as corticosteroids, thyroid dysfunction, obesity, diabetes, smoking, and high C-reactive protein value. This is problematic as, for example, cystatin C may not be useful in renal transplant patients because they have subclinical inflammation and commonly use long-term corticosteroids.

A meta-analysis published in 2002 showed that serum cystatin C measured with an immunonephelometric assay is more accurate than serum creatinine as a marker of GFR.^{4,5} However, as noted previously, the cystatin C value, like the creatinine value, may be affected by a number of factors other than the GFR.

Laboratory techniques to measure cystatin C include latex immunoassays, such as automated particle-enhanced turbidimetric immunoassay and nephelometric immunoassay. Other techniques are radioimmunoassay, fluorescent techniques, and enzymatic immunoassays. Most of these assays are more expensive than the measurement of serum creatinine and although the International Federation of Clinical Chemists has developed a reference material for standardization of cystatin C, international standardization remains in process.⁶⁻⁸

A number of equations utilizing cystatin C have been developed to estimate GFR. The 2008 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) cystatin C and CKD-EPI creatinine–cystatin C equations were developed in a preliminary study to assess the impact of cystatin C level on eGFR in individuals with CKD prior to the standardization of the cystatin C assay.⁹ The 2012 CKD-EPI cystatin C and the CKD-EPI cystatin C–creatinine equations were developed from a diverse group of 5352 individuals from 13 studies. These equations were then validated in 1119 participants from 5 different studies in which GFR had been measured. The creatinine–cystatin C equation was found to perform better than equations that used creatinine or cystatin C alone—it produced an eGFR that was within 20% of the measured GFR (mGFR) in a significantly higher proportion of participants. Bias with the new, combined creatinine–cystatin C equation and with the average of the new cystatin C equation and the creatinine equation was similar to that with the individual creatinine and cystatin C equations, but they had greater precision and accuracy and resulted in more accurate classification of GFR as < 60 mL/min/1.73 m², the threshold for the diagnosis of CKD.¹⁰ This was found to hold true even in individuals with low body mass index (a subgroup in which creatinine-based GFR estimates are known to be less accurate) and was less subject to the effects of age, sex, and race.

Other equations have been reported since the development of the CKD-EPI cystatin C and creatinine–cystatin C equations, which perform as well or better than the CKD-EPI cystatin C equation in some subgroups but do not perform better than the CKD-EPI equation in a diverse population.¹¹⁻¹⁴

It has been proposed that cystatin C–based equations may be more accurate in populations with lower creatinine production, such as in children, the elderly, or cirrhotic patients,¹⁵⁻¹⁷ although whether cystatin C correlates better with GFR than serum creatinine in patients with diabetic nephropathy is unclear.¹⁸ In addition, steroids may affect cystatin C levels, thus making its utility in transplant recipients unclear.¹⁵

The 2012 KDIGO guidelines recommend using serum creatinine and a GFR estimating equation for initial assessment and to use additional testing such as cystatin C or a clearance measurement for confirmatory testing if eGFR based on serum creatinine is less accurate. For reporting eGFR_{cys} and eGFR_{cr-cys} in adults, the guideline recommended using the 2012 CKD-EPI cystatin C equations or another cystatin C–based GFR estimating equation if it has demonstrated improved accuracy of GFR estimates compared with these CKD-EPI cystatin C equations. The guideline recommends measuring cystatin C in adults with eGFR_{cr} of 45–59 mL/min/1.73 m² without markers of kidney damage if CKD confirmation is required. If eGFR_{cys}/eGFR_{cr-cys} is also < 60 mL/min/1.73 m², the CKD diagnosis is confirmed. If eGFR_{cys}/eGFR_{cr-cys} is ≥ 60 mL/min/1.73 m², the CKD diagnosis is not confirmed.¹⁹ While this might help in risk stratification, the guidelines do leave a number of questions unanswered, such as how to incorporate cystatin-based GFR estimates into routine clinical practice and use this equation for longitudinal follow-up of patients. The NKF-KDOQI commentary on the KDIGO guideline concurred with their recommendations for GFR evaluation but recommended against widespread use of cystatin C because of concerns regarding incomplete understanding of its non-GFR determinants, higher costs, and incomplete standardization of assays.²⁰

Novel Endogenous Filtration Markers

Several alternative novel endogenous substances are under investigation as potential markers that could be used aside from urea, creatinine, and cystatin C. They are beta-trace protein (BTP), β_2 -microglobulin (B2M), and symmetrical dimethylarginine.

BTP is a 23–29-kDa protein, larger isoforms of which are found in the serum and urine. In late 2000s, the White and Pöge equations were developed, using BTP alone or in combination with urea or creatinine to determine eGFR. However, they were not felt to offer any significant benefit over the Modification of Diet in Renal Disease (MDRD) equation.^{21,22}

B2M is a 11.8-kDa protein commonly used as a prognostic marker for multiple myeloma. It is freely filtered by the glomerulus and is extensively reabsorbed and metabolized in the proximal tubule.²³ The CKD-EPI investigators attempted to develop eGFR equations using both BTP and B2M.²⁴ They found that although the combined equation was similar to the CKD-EPI creatinine and cystatin C–based equations, it did not offer an improvement over either, nor was it as accurate as the combined CKD-EPI equation.

Interestingly, both BTP and B2M were subsequently found to be strong predictors for CKD progression, cardiovascular disease, and mortality.^{25,26}

Used alone, these markers have a number of shortcomings, such as lack of standardization of assays, lack of certified reference materials, and a number of non-GFR determinants (such as male gender, markers of inflammation, and body

mass index). They do, however, hold promise as potential endogenous filtration markers that could not only improve GFR estimation in combination with other measurements (by reducing bias and improving accuracy), but also potentially predict the risk of ESKD and mortality. The future of GFR measurement will likely involve the use of kidney metabolomics, looking to use several plasma-based markers to give a better estimate of GFR, predict risk, and reduce the impact of individual non-GFR determinants.²⁷

Equations for Estimating GFR

Because direct measurement of GFR is not practical in clinical practice, estimation equations have been developed to aid clinicians in interpretation of serum creatinine, given the limitations previously cited.

A multitude of equations (Table 23.2) derived for the estimation of GFR are available. They all attempt to transform the serum concentration of a filtration marker into a value approximating the GFR, usually with the addition of other factors, such as age, sex, weight, and height, in part because the main filtration marker that is used is creatinine, which is known to be affected by these factors.

The main use of eGFR equations is in the detection, monitoring, and prognostication of CKD, which is largely asymptomatic. These equations have been derived from populations with stable CKD and are therefore not useful in the acute setting. Translating creatinine values, especially at the upper limit of normal (which as previously discussed do not reflect significant declines in GFR), into an eGFR has heightened the awareness of CKD. Estimation equations

Table 23.2 Most Commonly Used Equations for Estimating Glomerular Filtration Rate

Equation Name	Equation	Derivation Population
Cockcroft–Gault (1976)	$(140 - \text{age}) \times \text{wt (kg)} / \text{creatinine } (\mu\text{mol/L}) \times 0.81$ Female: $\times 0.85$	249 male veterans Median GFR 34
MDRD equation (1999)	$(140 - \text{age}) \times \text{lean body weight (kg)} / \text{Cr (mg/dL)} \times 72$	1628 patients enrolled in the MDRD Study (mean age, 50.6 years)
MDRD equation without ethnicity factor ^a	$175 \times \text{SCr}^{-1.154} \times \text{age}^{-0.26} \times 0.742$ (if female) $\times 1.212$ (if black)	Mean GFR 39.8 mL/min/1.73 m ²
CKD-EPI equation (2009)	$175 \times \text{SCr}^{2.190} \times \text{age}^{-0.26} \times 0.742$ (if female)	8254 participants from 6 research studies and 4 clinical populations (mean age, 47 years)
	$141 \times \min(\text{SCr}/\kappa, 1)^\alpha \times \max(\text{SCr}/\kappa, 1)^{-1.209} \times 0.993^{\text{Age}} \times 1.018$ (if female) $\times 1.159$ (if black), where κ is 0.7 for females and 0.9 for males α is -0.329 for females and -0.411 for males min indicates the minimum of SCr/κ or 1 max indicates the maximum of SCr/κ or 1	Mean GFR 68 mL/min/1.73 m ²
CKD-EPI cystatin C (2012)	$133 \times \min(\text{SCysC}/0.8, 1)^{-0.499} \times \max(\text{SCysC}/0.8, 1)^{-1.328} \times 0.996^{\text{Age}} \times 0.932$ (if female), where min indicates the minimum of $\text{SCysC}/0.8$ or 1 max indicates the maximum of $\text{SCysC}/0.8$ or 1	5352 participants from 13 studies (mean age, 47 years) Mean GFR 68 mL/min/1.73 m ²
CKD-EPI creatinine–cystatin C (2012)	$135 \times \min(\text{SCr}/\kappa, 1)^\alpha \times \max(\text{SCr}/\kappa, 1)^{-0.601} \times \min(\text{SCysC}/0.8, 1)^{-0.375} \times \max(\text{SCysC}/0.8, 1)^{-0.711} \times 0.995^{\text{Age}} \times 0.969$ (if female) $\times 1.08$ (if black), where α is -0.248 for females and -0.207 for males κ is 0.7 for females and 0.9 for males min($\text{SCr}/\kappa, 1$) indicates the minimum of SCr/κ or 1 and max($\text{SCr}/\kappa, 1$) indicates the maximum of SCr/κ or 1 min($\text{SCysC}/0.8, 1$) indicates the minimum of $\text{SCysC}/0.8$ or 1 and max($\text{SCysC}/0.8, 1$) indicates the maximum of $\text{SCysC}/0.8$ or 1	34 mL/min
BIS equation	$\text{BIS1} = 3736 \times \text{creatinine}^{-0.123} \times \text{age}^{-0.131} \times 0.82$ (if female)	610 individuals over 70 years of age
BIS1: creatinine	$\text{BIS2} = 767 \times \text{cystatin C}^{-0.98} \times \text{creatinine}^{-0.77} \times \text{age}^{-0.94} \times 0.87$ (if female)	Mean age: 78.5 years
BIS2: creatinine and cystatin C (2012)		
FAS equation: normalized serum creatinine (SCr/Q) (2016)	eGFR = $107.3 \times / [\text{Scr}/Q]$ for 2 years < age ≤ 40 years eGFR = $107.3 \times 0.988(\text{age} - 40)/[\text{Scr}/Q]$ for age >40 years	6870 individuals spanning all age groups (<18 to >70 years)

^aAfrican-American coefficient of the MDRD study equation.

BIS, Berlin Initiative Study; CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration; Cr, creatinine; eGFR, estimated glomerular filtration rate; FAS, full age spectrum; GFR, glomerular filtration rate; MDRD, Modification of Diet in Renal Disease; SCr, serum creatinine; SCr/Q, normalized serum creatinine, where Q is the median serum creatinine from healthy populations to account for age and gender; SCysC, serum cystatin C; wt, weight.

Adapted from Tables 12 and 16 in Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

also appear to be reasonably accurate for following changes in GFR over time.^{28,29}

Bias, Precision, and Accuracy. The performance of estimation equations is assessed by measurements of bias, precision, and accuracy.

- “Bias” results from the systematic underestimation or overestimation of the GFR in a population due to an error within the equation itself. It is calculated as the mean or median difference between the mGFR and eGFR values.
- “Precision” refers to the reliability and reproducibility of repeated measurements with one another. In the case of eGFR, it is usually expressed as the standard deviation or interquartile range of differences between eGFR and mGFR values.
- “Accuracy” combines bias and precision and is the most useful assessment of an estimation equation. It is measured as the percentage of GFR estimations within a particular percentage range (usually 30%) from their respective GFR measurements.

The problem of bias with eGFR equations can potentially be overcome with derivation of the equation from larger sample sizes and use of filtration markers such as cystatin C, which are less subject to interference by other factors. Accuracy, however, is difficult to achieve in a heterogeneous population. Even the latest Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation using creatinine, which is being adopted widely, has accuracy such that 80.6% of eGFR values are within 30% of mGFR (Fig. 23.2). This means that one in five values of eGFR in the general population is incorrect. Again, in day-to-day clinical practice, accuracy may not be necessary, and establishing an eGFR trend within an individual patient with CKD is probably more important.

Modification of Diet in Renal Disease and Chronic Kidney Disease Epidemiology Collaboration Equations. Although a number of GFR estimation equations have been developed, two equations are most widely used in clinical practice.

MDRD. The MDRD study equation was initially derived in 1999 from a population of 1628 individuals who had participated in the MDRD study, using creatinine measured by a modified Jaffe method.³⁰ A new MDRD equation was expressed in 2004 for IDMS-traceable creatinine values. The MDRD equation was compared against a gold standard, urinary clearance of iothalamate. The performance of the MDRD equation was evaluated in a number of populations because it was initially derived from a primarily Caucasian American population. A number of coefficients have been derived to compensate for differences in body mass and diet in populations of different ethnicity with varying degrees of performance.

The MDRD equation was widely adopted in the United States, Europe, and Australia, where eGFR is now routinely reported with serum creatinine results. It is gradually being replaced by the CKD-EPI equation.

The limitations of the MDRD equation are mainly its tendency to underestimate the GFR and its relatively low accuracy at higher GFR values.

CKD-EPI. The CKD-EPI equation was derived in 2009 in 8254 patients (a further 3896 patients pooled from 16 studies were used for validation), with urinary clearance of iothalamate again used as the gold standard.³¹ It has less bias and greater accuracy than the MDRD equation, especially at higher GFR. A meta-analysis involving 1.1 million adults showed that the CKD-EPI equation reclassified a significant number of patients (24.4%) into a higher GFR range³²; 34.7% of patients who were classified as having stage 3A CKD (eGFR 45–59 mL/min/1.73 m²) with the MDRD equation did not have GFRs < 60 mL/min/1.73 m² with the CKD-EPI equation

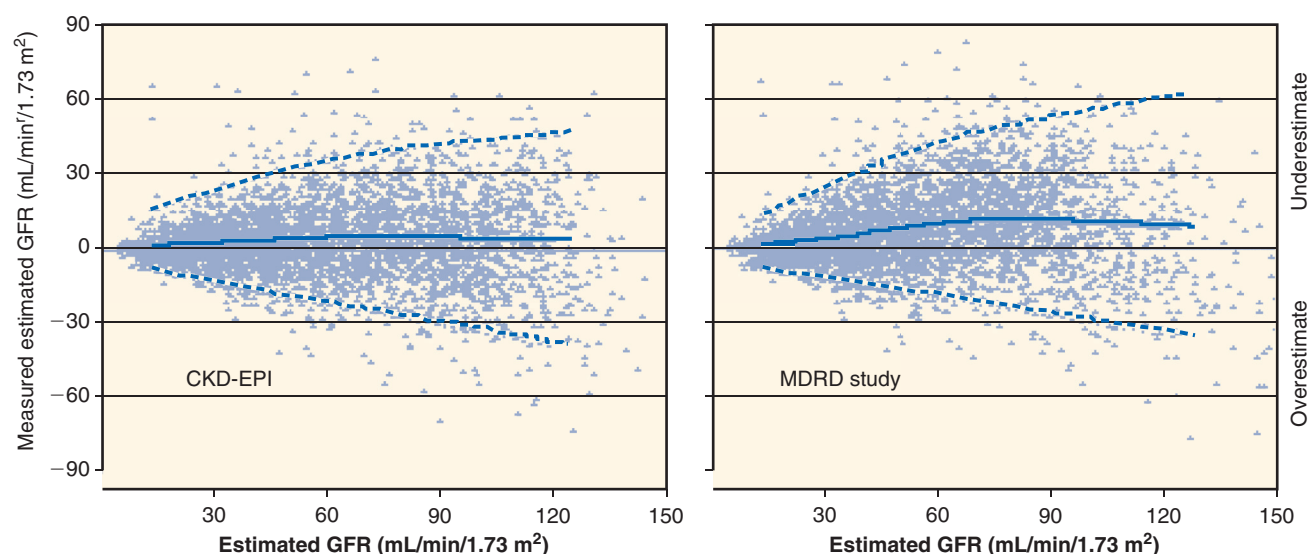


Fig. 23.2 Difference between measured and estimated glomerular filtration rate (GFR) using the Modification of Diet in Renal Disease (MDRD) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) estimation equations. Shown are smoothed regression lines and hashed 95% confidence interval lines. Although both the MDRD and CKD-EPI equations tend to underestimate the GFR, the CKD-EPI equation does so to a lesser degree, particularly at higher GFR. Therefore the CKD-EPI equation has less bias. (From Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med.* 2009;150:604–612.)

and were therefore no longer regarded as having CKD according to the newest definition. The precision of the CKD-EPI and MDRD equations is still suboptimal, however, because eGFR and mGFR values varied by at least 16.6 and 18.3 mL/min/1.73 m² (the interquartile range of differences for patients across the range of GFR), respectively.

CKD-EPI equations have also been derived for cystatin C and a combination of cystatin C and creatinine. Cystatin C measurement in the derivation of these equations was traceable to the standard reference material for cystatin C, although cystatin C measurement is not uniformly standardized, as noted previously. GFR estimation using cystatin C and creatinine in combination has been shown to be more accurate than either marker alone, but eGFR using cystatin C is not superior to eGFR using creatinine.³³ The KDIGO 2012 guidelines recommend initial GFR evaluation using serum creatinine and a GFR estimating equation and confirmatory testing using tests such as serum cystatin C or a clearance measurement in certain circumstances if eGFR_{cr} is less accurate. For confirmatory testing using a clearance measurement, the recommendations include using plasma or urinary clearance of an exogenous filtration marker rather than CrCl. Recommendations to clinical laboratories included using assays for creatinine and cystatin C that are traceable to reference standards, and reporting eGFR using the CKD-EPI equations, or alternative equations if they are superior.¹⁹

Risk Stratification of CKD with eGFR: eGFR equations transform the creatinine or cystatin C measurement into a value that approximates the mGFR. This value can also be used to classify the patient according to the stage of CKD. Decreasing values of GFR in patients with CKD have been associated with increasing risk for metabolic complications of CKD, end-stage kidney disease (ESKD), cardiovascular disease, and death.³⁴ The many eGFR equations have performed variably in their ability to predict certain outcomes in different populations. This difference in ability to risk stratify is due to the different degrees of importance placed on factors such as age and sex, which also affect prognosis, in these equations. The importance of accuracy in measuring GFR versus stratification of risk as a result of decreased GFR is a matter of ongoing debate and beyond the scope of this text. For further discussion of risk factors in CKD, please see Chapter 20.

Cockcroft–Gault Equation. The Cockcroft–Gault (CG) equation estimates CrCl rather than the GFR. This equation was developed in 1976 from a cohort of 249 men, and the creatinine assay method that was used to derive this equation was not standardized.³⁵ The equation is

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times \text{lean body weight (kg)}}{\text{Cr (mg/dL)} \times 72}$$

The CG equation has not been re-expressed since the adoption of new assay methods for estimation of creatinine. For these reasons, the CG equation systematically overestimates GFR and should not be used without an understanding of its limitations, although it is currently still often used to guide drug dosing including drug research. There is especially a risk of overdosing drugs that have a narrow therapeutic index with the CG equation.

EXOGENOUS GLOMERULAR FILTRATION MARKERS

Inulin

Inulin remains the gold standard filtration marker but its use is impractical for routine clinical purposes. Inulin is a polymer of fructose found in tubers such as the Jerusalem artichoke and chicory. It distributes in extracellular fluid, does not bind to plasma proteins, is freely filtered at the glomerulus, and is neither reabsorbed nor secreted by the renal tubules.

Inulin is intravenously infused at a constant rate while blood and urine are sampled frequently over several hours, ideally following insertion of a bladder catheter. The patient takes an oral water load and must continue consuming water throughout the test to ensure a high urine output.

Clearance Methods for Other Exogenous Glomerular Filtration Markers

Owing to the difficulty and expense of using inulin, new reference standard filtration markers have been introduced as alternatives and have been widely used since the 1990s. Urinary iothalamate and iothexol have now been adopted now been adopted as the reference standard for measurement of GFR. Clearance of these filtration markers can be measured in the urine or in blood or with nuclear imaging in the case of radiolabeled markers to avoid problems with urine collection.³⁶

GFR is calculated from plasma clearance after a bolus intravenous injection of an exogenous filtration marker, with clearance being calculated from the amount of the marker administered divided by the area under the curve of plasma concentration over time. The decline in serum concentration is due initially to the disappearance of the marker from the plasma into its volume of distribution (fast component) and then subsequently to renal excretion (slow component). It is best estimated using a two-compartment model that requires blood sampling early (usually two or three time points until 60 min) and late (one to three time points from 120 min onward). GFR can also be measured by counting a radioactive exogenous filtration marker over the kidneys and bladder areas; however, this technique is thought to be generally less accurate.

These methods are not perfect and are subject to imprecision, albeit at a much lower level than that of equations that estimate GFR.

Radiolabeled Markers

Radiolabeled iothalamate, ethylenediaminetetraacetic acid (EDTA), and diethylenetriaminepentaacetic acid (DTPA) are all used as glomerular filtration markers.

Iothalamate may be labeled with iodide I 125 (¹²⁵I) or used unlabeled. EDTA is commonly used in Europe, whereas DTPA is widely used in the United States. EDTA is usually labeled with chromium Cr 51 (⁵¹Cr), but EDTA may be reabsorbed by the tubules, leading to underestimation of GFR. DTPA is labeled with technetium Tc 99m (^{99m}Tc), and the major limitation of its use is the potential for an unpredictable dissociation of ^{99m}Tc from DTPA and binding to plasma proteins, resulting in underestimation of GFR.

Unlabeled Radiocontrast Agents

Because of concerns about exposure to radiation, storage, and disposal of radionuclide markers, techniques have been

developed to measure low levels of iodine in urine and plasma. Iothalamate and iothexol concentrations have been measured using HPLC, but the main disadvantage of this approach is the complexity of the HPLC assay. X-ray fluorescence of samples may also be used to measure iodine levels but requires a higher dose of contrast agent.

Iothexol is a nonionic, low-osmolar contrast agent. It is not reabsorbed, metabolized, or secreted by the kidney and is excreted completely unmetabolized in the urine. It has low toxicity and is usually used in doses 10 to 50 times higher in radiologic procedures than those used for GFR determination.

Urinary clearances of iothalamate and iothexol closely correlate with urinary inulin clearance, and are therefore the methods most commonly used to measure GFR.

SPECIFIC CIRCUMSTANCES OR POPULATIONS

CHILDREN

Adult GFR equations such as the MDRD have been shown to be inappropriate in children aged ≤ 9 years (see also Chapter 72). Multiple equations (Table 23.3) have been derived for estimation of GFR in children, but the most popular is the Schwartz equation, which was devised in 1976³⁷

$$eGFR = \frac{k \times L}{SCr}$$

where k depends on the age of the child, L is length or height, and SCr is serum creatinine. Like the CG equation, the Schwartz equation has now been shown to systematically overestimate GFR. The reasons are the creatinine assay technique and measurement factors specific to children. As

previously mentioned, the Jaffe method can be interfered with by plasma proteins, and the correction factor that is deducted to account for this interference is estimated by taking an average bias from measured results. In children, who have lower levels of plasma proteins, this correction factor may be too high, resulting in an erroneously low creatinine value. Because of the low muscle mass of children, the influence of a measurement error is also proportionately larger than an error of the same magnitude in an adult sample.

The simple bedside Schwartz equation— $eGFR = 0.413 \times (\text{height [cm]}/\text{serum creatinine [mg/dL]})$ —was developed using standardized creatinine methods in 2009.³⁸ This equation provides good approximation of a more complicated Schwartz eGFR formula, using creatinine, urea, and cystatin C as well as height.³⁸ Cystatin C has been suggested to be more accurate than creatinine as an indirect marker of renal function in children.³⁹

A limitation is that the equation was derived from a cohort of 600 children with CKD who had abnormal growth. Therefore this GFR estimation equation may not be accurate for children who have less impairment of renal function and normal skeletal growth (Table 23.3).

ELDERLY POPULATION

Elderly individuals were either not included or were poorly represented in the study populations used to develop the CG, MDRD, and CKD-EPI equations. The Berlin Initiative Study (BIS) attempted to develop an equation based on serum creatinine concentration, gender, and age using a study population that included 610 individuals aged >70 years. The equation was compared with a gold standard iothexol plasma clearance measurement, three creatinine-based equations

Table 23.3 Equations Using Serum Biomarkers for Estimating GFR in Children and Adolescents

		Derivation Population		
Equation Name	Equation	No. Children	Age (Years)	GFR Range or Median (mL/min)
Creatinine Based				
Schwartz, 1976	$0.55 \times \text{Ht}/\text{Scr}$	77	1–21	3–220
Counahan, 1976	$0.43 \times \text{Ht}/\text{Scr}$	103	0.2–14	4–200
Leger, 2002	$(0.641 \times \text{Wt})/\text{Scr} + (0.00131 \times \text{Ht}^3)/\text{Scr}$	97	1–21	97
Schwartz, 2009	$0.413 \times \text{Ht}/\text{Scr}$ $40.7 \times (\text{HT}/\text{Scr})^{0.640} \times (30/\text{BUN})^{0.25}$	349	1–17	41
Cystatin C Based				
Filler, 2003	$91.62 \times (\text{cysC})^{-1.123}$	85	1–18	103
Grubb, 2005	$84.69 \times (\text{cysC})^{-1.680} \times 1.384$ if <14 years	85	3–17	108
Zappitelli, 2006	$75.94 \times (\text{cysC})^{-1.17} \times 1.2$ if Tx	103	1–18	74
Creatinine and Cystatin C Based				
Bouvet, 2006	$63.2 \times (\text{Scr}/1.086)^{-0.19} \times (\text{cysC}/1.2)^{-0.93} \times (\text{Wt}/45)^{0.68} \times (\text{years}/14)^{0.77}$	100	1–23	92
Zappitelli, 2006	$43.82 \times e^{0.004} \times \text{Ht} \times (\text{cysC})^{-0.635} \times (\text{Scr})^{-0.547}$	103	1–18	74
Schwartz, 2009	$39.1 \times (\text{HT}/\text{Scr})^{0.516} \times (1.8/\text{cysC})^{0.294} \times (30/\text{BUN})^{0.40} \times (1.099)$ if male $\times (\text{HT}/1.4)^{0.9}$	349	1–17	41

BUN, Blood urea nitrogen; cysC, cystatin C; GFR, glomerular filtration rate; Ht/HT, height; Scr, serum creatinine; Tx, transplant; Wt, weight. Adapted from Table 3 in Schwartz GJ, Work DF. Measurement and estimation of GFR in children and adolescents. *Clin J Am Soc Nephrol.* 2009;4:1832–1843.

(CG, MDRD, and CKD-EPI), and three equations based on cystatin C and was shown to produce less bias than any of its counterparts. Four variables were taken into account—namely, age, sex, serum creatinine, and serum cystatin C—but not ethnicity as the cohort consisted of Germans of Caucasian origin. Two equations were derived—the BIS1 using serum creatinine only and BIS2 using serum creatinine and serum cystatin C (Table 23.2). These formulae showed better precision and agreement with mGFR, especially in a population with eGFR > 30 mL/min/1.73 m². The BIS equations appear to provide a more accurate estimation of GFR in the elderly and improve identification of individuals with CKD.¹¹ The BIS1 equation was subsequently shown to be useful in a Chinese group of 332 elderly subjects as well as in a different Caucasian cohort of 224 subjects.^{40,41}

PREGNANCY

It is well known that there is a physiologic increase in GFR, increased effective renal plasma flow, and a drop in serum creatinine occur in pregnancy. Serum creatinine decreases because of a real increase in GFR but also because of physiologic hemodilution.

Estimation equations are inappropriate for measuring GFR in pregnant women because they were not derived from this population. Most pregnant women have GFRs > 60 mL/min/1.73 m², which is above the range at which the MDRD or CKD-EPI equations are known to be accurate in any case. The 24-h urine collection for CrCl therefore remains the best method in pregnancy.⁴²

Nevertheless, eGFR has been used to study midterm renal hyperfiltration, a hemodynamic phenomenon that occurs early in pregnancy and persists until delivery.⁴³ This is regarded as a marker of renal functional reserve in pregnancy. Recently, Park et al. described a retrospective study looking at 1931 singleton pregnancies with midterm serum creatinine data between 2001 and 2015 in South Korea, using the CKD-EPI equation. Midterm hyperfiltration was defined as > 120 mL/min. They demonstrated a nonlinear U-shaped relationship between midterm eGFR and adverse outcomes, with values between 120 and 150 mL/min/1.73 m² being associated with the lowest incidence of adverse pregnancy outcomes (defined by preterm birth, low birth weight, or preeclampsia as well as the composite of these three features). There are several limitations to this study, but the data suggest that further research is needed to understand the pathophysiology of intrarenal hemodynamic dysfunction in pregnancy and how this may contribute to adverse outcomes.^{44,45}

ACUTE KIDNEY INJURY

Measurement of GFR in patients with AKI is difficult as it is constantly changing, and measurements or estimates that depend on a steady state are not readily applicable. Serum creatinine is the most commonly used marker of GFR in this setting, but creatinine is slow to rise in response to a decrease in GFR and is subject to the influence of dilution by fluid given as part of treatment for AKI.^{46,47} As previously mentioned, creatinine is also insensitive to substantial decreases in GFR. Estimation equations for GFR in patients with AKI are inaccurate because these equations have been derived from stable patients, in whom the creatinine is at a steady state, and are not applicable to the diverse AKI patient

population. A kinetic GFR estimation equation has been developed but has not yet been validated.⁴⁸

Accurate measurement of GFR in this population can be achieved by calculation of the elimination kinetics after a single bolus injection of a glomerular filtration marker. Alternatively, brief-duration measurements of GFR, such as short-duration urinary CrCl, can be performed, assuming that a patient's creatinine does not increase rapidly over 2 to 8 hours.⁴⁹ These tests are not widely performed and are probably impractical in many centers. This situation has led to extensive research into a biomarker that is more sensitive in demonstrating early stages of AKI to facilitate more timely intervention.

DRUG DOSING

The CG equation, despite its limitations, is still used regularly in many jurisdictions for drug dose adjustment in patients with renal impairment. Historically this equation has been used to enroll subjects in renal impairment categories for pharmacokinetic studies. The U.S. Food and Drug Administration (FDA) in its “Guidance for Industry Pharmacokinetics in Patients with Impaired Renal Function” suggested using either the MDRD equation or the CG equation for this purpose in new drug development,⁵⁰ although one should be cautious as this does not imply that CrCl is equivalent to eGFR. The FDA guidance has not been updated since the KDIGO and KDOQI have recommended that the CKD-EPI equation replace the MDRD equation in routine clinical practice. Because the CKD-EPI and MDRD study equations perform similarly at GFR < 60 mL/min/1.73 m², it would be reasonable to use eGFR computed using the CKD-EPI equation or the MDRD study equation for pharmacokinetic studies and drug dosing.⁵¹

A large simulation study comparing drug dosages administered to patients through the use of the MDRD and CG equations showed that the concordance figures of the two equations' GFR estimates with mGFR was similar.⁵² Because the majority of FDA-approved drug dosing labels use the CG equation, which expresses CrCl in mL/min, the eGFR value corrected for BSA must be converted to units of mL/min by multiplying by the individual's BSA and dividing by 1.73 m². The simulation study concluded that either equation can be used for drug dosing but that caution should be exercised in patients in whom the creatinine value may be inaccurate. This caution is particularly relevant in sick or hospitalized patients, in whom low body weight or changes in body weight are present, in the elderly, and in amputees. The CG equation in these cases tends to give a lower estimate of renal function,⁵³ leading to a dose reduction.⁵⁴ The lower estimate results from the greater effect of age and weight on the CG equation. The use of the CG equation may lead to drug dosing errors that are due to “underdosing,” thus minimizing exposure to toxicity. By contrast, in the elderly patients, the MDRD and CKD-EPI equations have been shown to overestimate the GFR in some studies but to be reliable in others.^{55,56} The elderly patients are more likely to experience side effects and to be subject to the dangers of polypharmacy, so using the CG equation in these patients may be more appropriate.⁵³

Ultimately the importance of drug dosage adjustment depends on the purpose of the medication as well as the therapeutic range and toxicity of the drug. In cases in which a medication has a narrow therapeutic window, such as

chemotherapy, all estimation equations may have an unacceptable degree of error, and accurate measurement of GFR using an exogenous filtration marker should be performed.

The CG equation should not be completely abandoned in favor of the MDRD equation, especially because recommendations for drug dosages of existing medications were based on the CG equation. The CKD-EPI equation has not yet been considered in this context.

FULL AGE SPECTRUM METHOD OF EGFR ESTIMATION

Owing to the methods by which various equations were developed for different populations, there can be a discontinuity when switching from pediatric equations to adult equations or from adult to older-adult equations. The full age spectrum (FAS) equation, developed in 2016, tries to correct for this disparity by deriving a single (age-knotted) eGFR equation that is applicable to all ages by normalizing serum creatinine for age (for children and adolescents) and sex (for adolescents and adults). A normalized serum creatinine (SCr/Q) is used, where Q is the median serum creatinine from healthy populations to account for age and gender (see Table 23.2).

Validation was performed in a total of 6870 healthy and kidney disease-affected white individuals including 735 children (<18 years of age), 4371 adults (aged between 18 and 70 years), and 1764 older adults (>70 years of age) with mGFR (inulin, iothexol, and iothalamate clearance) and IDMS-equivalent serum creatinine. Bias, precision, and accuracy were tested. The equation was found to be less biased and more accurate than the Schwartz equation for children and adolescents, less biased and as accurate as the CKD-EPI for young and middle-aged adults, and more accurate and less biased than CKD-EPI for older adults. The equation may not need further changes for age and gender but may need validation in other ethnicities and derivation of Q values for different populations. External validation in a large population suggested that the FAS equation might perform better than the Schwartz and CKD-EPI equations. There was better accuracy and precision for older adults with mGFR > 60 mL/min/1.73 m². As the FAS is based on SCr/Q values for a healthy population, it is expected that this will work better for a healthy and general population rather than in patients with CKD; however, in the validation population subgroups with mGFR < 60 mL/min/1.73 m² the FAS equation was not found to perform worse than the CKD-EPI equation.⁵⁷

RENAL TRANSPLANT RECIPIENTS

Currently, renal transplant function is monitored using the endogenous marker serum creatinine along with eGFR. Serum creatinine, however, is affected by a number of drugs used in transplant medicine. For instance, corticosteroids are shown to increase eGFR in animal models as well as in humans; glomerular micropuncture studies have shown that corticosteroids vasodilate both the afferent and efferent arterioles, resulting in increased plasma flow and hence a higher eGFR. Chronic steroid administration has also been shown to increase prostaglandin synthesis in some species.⁵⁸ Paradoxically, steroids also increase serum creatinine by about 10%. This is presumably due to steroid-induced hypercatabolic state with associated protein wasting and muscle loss. In addition, muscle mass in transplant recipients may differ

from the general population and those with other clinical conditions due to a chronic inflammatory state from chronic illness, infection, and episodes of rejection.^{59–61} In addition, trimethoprim used for *Pneumocystis jirovecii* prophylaxis can cause elevated creatinine by blocking its tubular secretion. As discussed earlier, eGFR can decrease substantially before there is an evident increase in serum creatinine levels; thus calcineurin inhibitor toxicity and rejection may not be detected early enough.

Serum creatinine tends to be used in renal transplant recipients to detect rejection, infection, or drug toxicity and eGFR is infrequently used to guide management decisions due to concerns that the equations may be less reliable in transplant recipients than other populations. Nevertheless, in a systematic review of 3622 solid-organ transplant recipients, Shaffi et al. demonstrated that the CKD-EPI and the four-variable MDRD study equations were more accurate than alternative equations (even those developed in populations of transplant recipients only) and as accurate as observed in other clinical populations. CKD-EPI performed better at higher GFRs and MDRD performed better at lower GFRs. However, these equations were still inaccurate (different from mGFR by more than 30%) in one of five patients.^{62,63}

URINALYSIS

Urinalysis may be used in the assessment of acute kidney disease or CKD, workup for kidney stones, or the evaluation of systemic conditions with potential renal involvement, such as systemic lupus erythematosus. There are three ways to obtain a urine specimen: spontaneous voiding, urethral catheterization, and percutaneous bladder puncture. Technique is important in collecting a sample to avoid contamination. For spontaneously voided urine, a midstream sample should be collected after cleaning of the external genitalia. If a patient has an indwelling catheter, a fresh specimen should be submitted for analysis; samples that have been stagnant in the catheter tubing or bag may have undergone degradation. Suprapubic needle aspiration of the bladder is used when urine cannot easily be obtained by other means, most commonly in infants. Whatever the collection method, it is recommended that a sample be analyzed within 2 to 4 hours of the time of collection to prevent cell lysis and precipitation of solutes.⁶⁴ There are numerous techniques for examining urine; this section focuses on methods commonly used for assessment of chemical content and microscopy.

COLOR

The color of urine is determined by chemical content, concentration, and pH (Table 23.4). Urine may be almost colorless if the output is high and the osmolality is low. Abnormal color changes can be due to drugs, foods, and pathologic conditions. Cloudy urine is most commonly due to leukocytes and bacteria. The most common cause of red urine is hemoglobin. Red urine in the absence of red blood cells in the sediment usually indicates either free hemoglobin or myoglobin. In the latter case, the patient's serum is not pink. Red urine with red sediment indicates hemoglobin. By contrast, red urine with clear sediment is most often the result of myoglobin but may also be seen in some porphyrias,

Table 23.4 Main Causes of Abnormal Color Changes in Urine

	Cause	Color
Pathologic conditions	Gross hematuria, hemoglobinuria, myoglobinuria	Pink, red, brown, black
	Jaundice	Yellow to brown
	Chyluria	White milky urine
	Massive uric acid crystalluria	Pink
	Porphyria, alkaptonuria	Red to black; increases after urine left to stand
Medications	Rifampin	Yellow-orange to red
	Propofol	White
	Phenytoin, phenazopyridine	Red
	Chloroquine, nitrofurantoin	Brown
	Triamterene, blue dyes of enteral feeds	Green
	Metronidazole, methyldopa, imipenem-cilastatin	Darkening after urine left to stand
Foods	Beetroot	Red
	Senna rhubarb	Yellow to brown red

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067; and Davison A. *Urinalysis*. 3rd ed. Oxford: Oxford University Press; 2005.

or with the use of some medications or the ingestion of beets in some individuals.⁶⁵

ODOR

The most common cause of abnormal urine odor is infection and in this case the abnormal odor is caused by the production of ammonia by bacteria. Ketones may cause a fruity or sweet odor. Some rare pathologic conditions may confer a specific odor to the urine. Examples are maple syrup urine disease (maple syrup odor), phenylketonuria (mousy odor), isovaleric acidemia (sweaty feet odor), and hypermethioninemia (fishy odor).⁶⁴

RELATIVE DENSITY

The concentration, or relative density, of urine can be assessed by either specific gravity or osmolality. “Specific gravity” is defined as the weight of a solution relative to that of an equal volume of water. It is determined by the number and size of particles in the urine. Specific gravity is traditionally measured by a urinometer, which is a weighted float marked with a scale from 1.000 to 1.060. This method is simple but outdated owing to the need for a larger volume of urine than with other methods and the potential for inaccuracy in reading the device. Today, specific gravity is commonly measured by refractometry or dry chemistry methods. Refractometry measures specific gravity using the refractive index of a solution, which is a function of the weight of solute per unit volume. It requires only a drop of urine.

Dry chemistry techniques are used in reagent strips. An indirect method is used to determine specific gravity, relying on the fact that there is generally a linear relationship between urine’s ionic strength and its specific gravity. The reagent strip contains a polyionic polymer that has binding sites that are saturated with hydrogen ions, and an indicator substance. The release of hydrogen ions when they are competitively replaced with urinary cations causes a change in the pH-sensitive indicator dye. Specific gravity values measured by dipstick tend to be falsely high if the urine pH is <6 and

falsely low if the pH is >7. The effects of nonionized molecules such as glucose and urea on osmolality are not reflected by changes in the dipstick specific gravity. Dry chemistry measurements of specific gravity therefore tend to correlate poorly with refractometry and osmolality.⁶⁴

“Osmolality,” the gold standard for relative density, is defined as the number of osmoles of solute per kilogram of solvent. It is measured directly with an osmometer. It depends on the number of particles in solution and is not influenced by their size or temperature. High-glucose solutions significantly increase osmolality (10 g/L glucose = 55.5 mOsm/L).⁶⁴ Urine specific gravity is generally proportional to the osmolality and rises by approximately 0.001 for every 35- to 40-mOsm/kg increase in urine osmolality.⁶⁶ A urine osmolality of 280 mOsm/kg (which is isosmotic to normal plasma) is usually associated with a urine specific gravity of 1.008 or 1.009. Specific gravity is affected by protein, mannitol, dextrans, and radiographic contrast media. In these settings, specific gravity can be increased disproportionately to the osmolality, falsely suggesting highly concentrated urine. There are no causes of a falsely low urine specific gravity value, and thus a specific gravity ≤1.003 measured by refractometry always indicates a maximally dilute urine (≤100 mOsm/kg).

URINE PH

Urine pH is usually measured with a reagent test strip. Most commonly, the double indicators methyl red and bromothymol blue are used in the reagent strips to give a broad range of colors at different pH values. The normal range for urine pH is 4.5 to 7.8. Significant deviations from true pH occur with values <5.5 or >7.5 with reagent strip methods.^{52,56} Urine pH can be useful in diagnosing systemic acid–base disorders when used in conjunction with other investigations, although in isolation it provides little useful diagnostic information. A very alkaline urine (pH > 7.0) is suggestive of infection with a urea-splitting organism, such as *Proteus mirabilis*. Prolonged storage can lead to overgrowth of urea-splitting bacteria and the laboratory measurement of a high urine pH. Diuretic therapy, vomiting, gastric suction, and alkali

therapy can also cause a high urine pH. Acidic urine (pH < 5.0) is seen most commonly in metabolic acidosis. A urine pH > 5 in the setting of metabolic acidosis may indicate one of the forms of renal tubular acidosis, though there are forms of renal acidosis in which the urine pH is low despite a defect in the total kidney ability to excrete acid and generate bicarbonate (see also Chapters 16 and 24).⁶⁵

BILIRUBIN AND UROBILINOGEN

Only conjugated bilirubin is passed into the urine. Thus the result of a reagent test for bilirubin is typically positive in patients with obstructive or hepatocellular jaundice, whereas it is usually negative in patients with jaundice due to hemolysis. In patients with hemolysis, the urine urobilinogen result is often positive. Reagent test strips are very sensitive to bilirubin, detecting as little as 0.05 mg/dL. However, the measurement of bilirubin in the urine is not very sensitive for detecting liver disease. Prolonged storage and exposure to light can lead to false-negative results.⁶⁵ False-positive test results for urine bilirubin can occur if the urine is contaminated with stool.

LEUKOCYTE ESTERASE AND NITRITES

The esterase method relies on the fact that esterases are released from lysed urine granulocytes (leukocytes). These esterases liberate 3-hydroxy-5-phenylpyrrole after substrate hydrolysis. The pyrrole reacts with a diazonium salt in the test strip, yielding a pink to purple color. The result is usually interpreted as negative, trace, small, moderate, or large. Factors that may increase leukocyte lysis include allowing urine to stand for long periods, low pH, and low relative density. In these settings, there may be a positive dipstick result for leukocyte esterase with no leukocytes seen on microscopy. High levels of glucose, albumin, ascorbic acid, tetracycline, cephalexin, or cephalothin or large amounts of oxalic acid may inhibit the reaction and cause false-negative results.⁶⁷

Urinary nitrites indicate the presence of nitrate-reducing bacteria. In the reagent strip test, nitrite reacts with a *p*-arsanilic acid to form a diazonium compound; further reaction with 1,2,3,4-tetrahydrobenzo(h)quinolin-3-ol results in a pink color endpoint. Results are usually interpreted as positive or negative. High specific gravity and ascorbic acid may interfere with the test. False-negative results are common and may be due to prolonged sample storage or low dietary intake. It may take up to 4 hours to convert nitrate to nitrite, so inadequate bladder retention time can also give false-negative results.⁶⁷

GLUCOSE

Glycosuria due to hyperglycemia may occur at blood glucose levels >10 mmol/L (180 mg/dL) in subjects with normal renal function. Less commonly, glycosuria indicates failure of proximal renal tubular reabsorption in tubular disorders such as Fanconi syndrome. Most reagent strips use an oxidase–peroxidase method to measure glucose. Glucose is first oxidized to form glucuronic acid and hydrogen peroxide. Hydrogen peroxide then reacts via a peroxidase with a reduced chromogen to form a colored product.⁶⁴ This test is

sensitive to glucose concentrations between 0.5 and 20 g/L.⁶⁴ Large quantities of ketones, ascorbate, and phenazopyridine hydrochloride (Pyridium) metabolites may interfere with the color reaction, causing false-negative results. Oxidizing agents and hydrochloric acid may cause false-positive results.⁶⁷ Enzymatic methods such as a hexokinase give more precise quantification of urinary glucose levels.⁶⁴

KETONES

Ketones (acetoacetate and acetone) are generally detected with the nitroprusside reaction.⁶⁷ Ascorbic acid and phenazopyridine can give false-positive reactions. It is important to appreciate that β -hydroxybutyrate (often 80% of total serum ketones in ketosis) is not normally detected by the nitroprusside reaction. Ketones may appear in the urine, but not in serum, with prolonged fasting or starvation. Ketones may also be detected in the urine in alcoholic or diabetic ketoacidosis.

HEMOGLOBIN AND MYOGLOBIN

Reagent strips use the peroxidase-like activity of the heme moiety of hemoglobin to catalyze the reaction between a peroxide and a chromogen, giving a colored product. This test is very sensitive for the presence of heme in the urine. False-negative results are uncommon but may be caused by ascorbic acid, a strong reducing agent. False-positive results may occur because of oxidizing contaminants, povidone-iodine, semen, or a high concentration of bacteria with pseudoperoxidase activity (such as Enterobacteriaceae, staphylococci, and *Streptococcus* spp).⁶⁴ Normally, haptoglobin binds circulating heme-containing substances, such as hemoglobin and myoglobin. When these substances are produced in large quantities, as occurs in hemolysis or rhabdomyolysis, the capacity for binding is overwhelmed and they appear in the urine. A positive dipstick test result for hemoglobin in the absence of red blood cells in the urine sediment therefore suggests either hemolysis or rhabdomyolysis.

PROTEINURIA

Proteinuria (see also Chapter 30) is an important sign of kidney disease, imparting powerful diagnostic and prognostic information. It is a cornerstone of the workup for CKD, AKI, hematuria, and preeclampsia. It is often the earliest marker of glomerular diseases, occurring before a reduction in GFR. Proteinuria is associated with hypertension, obesity, and vascular disease. It can be used to predict risks of CKD progression, cardiovascular disease, and all-cause mortality in general population⁶⁸ cohorts and patients with diabetes⁶⁹ and CKD.⁷⁰ Proteinuria-lowering therapies may be renoprotective,⁷¹ and monitoring proteinuria is a key aspect of assessing treatment response in a variety of kidney diseases, including diabetes⁷² and nondiabetic glomerulopathies.¹⁹ In addition, filtered protein probably contributes to the pathogenesis of renal injury and disease progression rather than just being a marker of it.⁷³ Although measurement of urinary protein has long been recommended in clinical practice guidelines, recommendations regarding this practice vary substantially.⁷⁴ This section reviews the normal physiology of proteinuria as well

as strengths, limitations, and applications of the different measurement techniques.

NORMAL PHYSIOLOGY

In humans, on the basis of a GFR of 100 mL/min, 180 L of primary urine is produced per day from plasma that contains about 10 kg of protein. However, only about 0.01% or 1 g of protein passes through the glomerular filtration barrier into the filtrate.⁷⁵ The glomerular filtration barrier acts as a size-, shape-, and charge-dependent permselective molecular sieve, the unique properties of which are still incompletely understood. It restricts the passage of macromolecules, such as albumin and globulin, and enables the excretion of an almost protein-free ultrafiltrate containing water and small solvents.⁷⁶

The glomerular filtration barrier acts to minimize diffusion of large molecules (with a Stokes–Einstein radius >1.5 nm)⁷⁷ that would otherwise occur down a concentration gradient from the plasma to the filtrate (see also Chapter 3). It is composed of three major layers—endothelial cells, the glomerular basement membrane, and podocytes, which cover the basement membrane on the side of the urinary space. Podocytes are highly specialized epithelial cells with long, interdigitated foot processes that wrap around the glomerular capillaries, forming 40-nm wide gaps, known as filtration slits, between adjacent processes (see also Chapter 4).⁷⁸ The slit diaphragm is a cell-to-cell contact that inserts laterally into the podocyte cell membrane, bridging the filtration slit. The podocyte plays a central role in integrating the components of the glomerular filtration barrier by interacting with the glomerular basement membrane and signaling at the slit diaphragm.⁷⁹ To date, at least 26 podocyte-specific gene defects, such as those encoding for the podocyte proteins nephrin and podocin, have been identified in hereditary causes of nephrotic syndrome.^{80–82} In response to signals from the podocytes and mesangium, the endothelial cells acquire a highly fenestrated phenotype, with small pores covering about 20% of their surfaces.⁷⁸ This phenotype facilitates high-flux transport of fluid and small solutes. Normally, large quantities of high-molecular-weight (HMW) plasma proteins traverse the glomerular capillaries, mesangium, or both without entering the urinary space. Damage to any one of the three layers of the glomerular filtration barrier allows proteins through, resulting in abnormal, “glomerular” proteinuria.

Albumin, the dominant HMW protein in plasma, is a negatively charged, approximately 67-kDa protein.⁷⁶ Size selectivity restricts the passage of albumin through the glomerular filtration barrier. Charge selectivity, in which the negatively charged proteoglycans and heparan sulfates in the glomerular basement membrane repel albumin molecules, is a theory seeking to explain the low glomerular sieving coefficient of albumin in relation to other molecules of its size.⁸³ However, experimental data have called the role of basement membrane charge in permselectivity into question.^{84–86} Some albumin filtration across the capillary wall does occur, after which it is resorbed by the proximal tubule cells.^{87,88}

LMW proteins ($<20,000$ Da) pass readily across the glomerular capillary wall. Because the plasma concentration of these proteins is much lower than that of albumin and globulins, however, the filtered load is small. Moreover, LMW proteins are normally reabsorbed by the proximal tubule.

Thus proteins such as α_2 -microglobulin, apoproteins, enzymes, and peptide hormones are normally excreted in only very small amounts in the urine.⁶⁵

A small amount of protein that normally appears in the urine is the result of normal tubular secretion. Tamm–Horsfall protein is an HMW glycoprotein (23×10^6 Da) that is formed on the epithelial surfaces of the thick ascending limb of the loop of Henle and early distal convoluted tubule. Immunoglobulin A (IgA) and urokinase are also secreted by the renal tubule and appear in the urine in small amounts.⁶⁵

TYPES OF PROTEINURIA

The types of proteinuria are as follows:

Glomerular: Increased filtration of macromolecules across the glomerular filtration barrier may occur from a loss of charge and size selectivity. Unlike the other types listed here, glomerular proteinuria often results in urinary protein loss >1 g/day.

Tubular: Tubular damage or dysfunction may inhibit the normal resorptive capacity of the proximal tubule, resulting in higher amounts of mostly LMW proteins in the urine. A degree of tubular proteinuria often occurs with glomerular proteinuria. Classic causes of tubular proteinuria in isolation are Fanconi syndrome and Dent disease.

Overflow: Normal or abnormal plasma proteins produced in increased amounts may be filtered at the glomerulus and may overwhelm the resorptive capacity of the proximal tubule. This occurs particularly with small or positively charged proteins and is of clinical importance principally in plasma cell dyscrasias (e.g., myeloma). It may also occur with myoglobin in rhabdomyolysis and with hemoglobin in severe intravascular hemolysis.

Postrenal: Small amounts of protein, usually nonalbumin IgG or IgA, may be excreted in the urinary tract in the setting of infection or stones. Leukocytes are also commonly present in the urine sediment.

NORMAL LEVELS OF PROTEINURIA

As previously described, two main groups of proteins are present in the urine: plasma proteins, predominately albumin, that cross the filtration barrier, and nonplasma proteins, predominately Tamm–Horsfall protein, that originate in renal tubules or the urinary tract. In normal physiologic conditions, about half of the excreted protein is Tamm–Horsfall protein, and <30 mg of albumin is excreted per day.⁷⁴ At normal levels of protein loss, albumin accounts for approximately 20% of total protein. As protein loss increases, albumin becomes the most significant single protein present.⁸⁹

CATEGORIZATION OF PROTEINURIA

The persistent excretion of abnormal levels of urinary albumin, equivalent to between 30 and 300 mg/day, that is below the level that can be detected by a standard urine protein dipstick is now termed “moderately increased albuminuria” (historically termed “microalbuminuria”) and corresponds to the KDIGO albuminuria category A2. Albumin excretion of >300 mg/day is now considered as “overt

Table 23.5 Kidney Disease: Improving Global Outcomes (KDIGO) Guideline: Categories of Proteinuria^a

	Normal to Mildly Increased (KDIGO A1)	Moderately Increased (KDIGO A2)	Severely Increased (KDIGO A3)
AER (mg/24 h)	<30	30–300	>300
PER (mg/24 h)	<150	150–500	>500
ACR:			
mg/mmol	<3	3–30	>30
mg/g	<30	30–300	>300
PCR:			
mg/mmol	<15	15–50	>50
mg/g	<150	150–500	>500
Protein reagent strip	Negative to trace	Trace to +	+ or greater

^aRelationships between AER and ACR and between PER and PCR are based on the assumption that average creatinine excretion rate is 1.0 g/day or 10 mmol/day (conversions are rounded for pragmatic reasons).

ACR, Albumin-to-creatinine ratio; AER, albumin excretion rate; PCR, protein-to-creatinine ratio; PER, protein excretion rate.

From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

proteinuria or severely increased albuminuria” (KDIGO albuminuria category A3, historically termed “macroalbuminuria”) and can be detected by a standard urine dipstick. “Nephrotic-range proteinuria” is protein excretion of >3.5 g/24 hours and is usually indicative of glomerular pathology. The 2012 KDIGO guidelines discourage the use of the term “microalbuminuria,” instead suggesting that the term “albuminuria” be used and the level subsequently quantified (Table 23.5).¹⁹ This recommendation has been sanctioned by the Association of Laboratory Physicians and Clinical Chemists in different jurisdictions. The presence of proteinuria or albuminuria strongly predicts outcomes of CKD progression as well as of cardiovascular and all-cause mortality in the population with CKD. The risk rises continuously as albuminuria increases.⁹⁰ The 2012 KDIGO guidelines added a category for albuminuria to the classification of CKD to improve risk stratification.¹⁹

SOURCES OF ERROR IN MEASUREMENT

The 24-hour urine collection for protein measurement is considered the gold standard for measuring protein or albumin (Tables 23.6 and 23.7). Methods such as reagent strips, random measurement of protein or albumin concentrations, and albumin- or protein-to-creatinine ratios (ACRs or PCRs) aim to estimate a 24-hour protein measure. A positive result from a semiquantitative test, such as a urinary dipstick test, should prompt further evaluation with a quantitative test. Both preanalytical factors and factors intrinsic to the analysis itself can be sources of error in protein measurement. In the assessment of the quality of a test, both accuracy and precision need to be taken into account. Although a test may give reproducible results, it may not accurately measure all clinically significant types of proteinuria. The heterogeneous types of protein and the different molecular forms of proteins (such as albumin) that may be present in urine make for a challenge to both accuracy and precision of measurement. Using a consistent form of measurement with a consistent assay to monitor proteinuria, and using multiple measurements to confirm findings, is therefore advisable.

Table 23.6 Patient Factors That May Increase Urinary Protein or Albumin

Posture (postural proteinuria)
Urinary tract infection
Hematuria
High dietary protein intake
High-intensity exercise
Congestive cardiac failure
Menstruation or vaginal discharge
Drugs (e.g., nonsteroidal antiinflammatory drugs)

Adapted from Johnson DW, Jones GR, Mathew TH, et al. Chronic kidney disease and measurement of albuminuria or proteinuria: a position statement. *Med J Aust.* 2012;197:224–225; and Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion [article in French]. *Ann Biol Clin (Paris).* 2010;68:9–25.

Table 23.7 Factors Influencing Accuracy of Proteinuria Measurement

Preanalysis Phase	Analysis Phase
Collection type Timed or random timing of random measurement	Total protein or albumin measurement
Degradation of protein or albumin during storage adsorption to plastic Storage temperature	Assay type

Adapted from Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion [article in French]. *Ann Biol Clin (Paris).* 2010;68:9–25; and Martin H. Laboratory measurement of urine albumin and urine total protein in screening for proteinuria in chronic kidney disease. *Clin Biochem Rev.* 2011;32:97–102.

ADVANTAGES OF URINARY ALBUMIN OVER TOTAL PROTEIN MEASUREMENTS

Either albumin or total protein can be measured in urine. Many current guidelines recommend the measurement of urine albumin on the basis of a need to detect lower levels of protein than were previously thought to be clinically significant. Multiple studies have shown that the presence of small amounts of albumin in the urine—between 30 and 300 mg/day—have prognostic significance; increasing amounts of albuminuria are associated with continuous increases in risk of all-cause and cardiovascular mortality, AKI, and ESKD in the general population,⁹¹ with decline in eGFR,^{92,93} and with adverse outcomes in CKD.⁹⁴ Measures of total protein are imprecise at low levels of protein and are insensitive at detecting clinically important changes in albuminuria.⁷⁴ Relatively large increases in urinary albumin excretion can occur without causing a measurable increase in urinary total protein.⁹⁵ Urine albumin measurements are more specific and more sensitive for changes in glomerular permeability than are measures of urinary total protein.^{19,74,95–97} In addition, because a single protein is being measured, standardization of albumin measurement is simpler than standardization of total protein measurements.⁹⁸

CONSIDERATIONS REGARDING MEASURING URINARY ALBUMIN RATHER THAN TOTAL PROTEIN

EVIDENCE FOR CKD PROGRESSION RISK AND INTERVENTIONS

Much evidence on the natural history and progression of CKD has centered on measurement of 24-hour total protein.⁹⁸ In general, studies of diabetic patients have used measurements of urinary albumin, but studies of interventions and outcomes for glomerular diseases, preeclampsia, and in children have used proteinuria measurements. One difficulty with the implementation of albumin as a replacement for total protein is the lack of a constant numerical relationship between the two that would enable clinicians to translate the existing evidence base from one to the other.⁷⁴

MISSED TUBULAR PROTEINURIA

Relying on measurement of urinary albumin risks missing “tubular” and “overflow” proteinuria, in which nonalbumin proteins predominate. However, total protein assays are generally more sensitive to albumin than to LMW proteins, and many also have poor sensitivity for detecting tubular proteinuria.⁹⁹ Although “tubular” disorders are characterized by a relative increase in the proportion of LMW protein to albumin, albumin generally still constitutes a significant portion of total protein, probably because of failure of tubular resorption of protein.¹⁰⁰ In the AusDiab study, random urine samples from >10,000 people in the Australian adult population were tested, using cutoff values of 3.45 mg/mmol for ACR and 22.6 mg/mmol for PCR. Of patients who screened positive for albuminuria, 68% had negative results for proteinuria. Albuminuria performed well as a screening test for proteinuria: sensitivity was 91.7%, specificity 95.3%, and negative predictive value 99.8%. However, among those with proteinuria, 8% excreted albumin within the normal range. The investigators postulated that these individuals may have

had light-chain proteinuria or interstitial nephropathies.⁸⁹ In a study of 23 patients with Dent disease, a rare but classic tubular disorder, only 2 patients had no significant urinary albumin loss in addition to losses of LMW proteins. In these two patients, the levels of LMW proteinuria were low enough that they would probably also have been missed by a total protein measurement approach.¹⁰¹

Overall, the significance of this issue in both the CKD and general populations is difficult to estimate with the currently available data. If tubular proteinuria is suspected, it is best assessed with immunoassays directed at a specific tubular protein, such as α_1 -microglobulin or monoclonal heavy or light chains.^{19,74} One study that attempted to identify the proteins composing tubular proteinuria in elderly people with mild proteinuria was unable to consistently identify proteins using electrophoresis, and the researchers suggested that the elevated urinary protein measurements were due to artifact.¹⁰²

DIAGNOSTIC UTILITY OF PROTEIN TYPE

Simultaneous measurement of different types of urinary protein may be a useful tool in differentiating between glomerular and tubulointerstitial diseases. Several studies using gel electrophoretic techniques to separate proteins on the basis of molecular size have shown that larger proteins such as albumin predominate in glomerular disease and that the ratio of LMW proteins is increased in tubulointerstitial disorders.¹⁰³ Higher albumin-to-total protein ratios, obtained from simultaneous measurement of ACR and PCR, have been shown to be significantly associated with glomerular rather than nonglomerular pathology on renal biopsy in patients with kidney disease.¹⁰⁴

METHODS TO MEASURE URINARY TOTAL PROTEIN

Total protein in urine has been measured by chemical, turbidimetric, and dye binding (colorimetric) methods (Table 23.8). These methods are prone to interference by inorganic ions and nonprotein substances in the urine.⁷⁴ Falsely high results may occur as result of interference by aminoglycoside antibiotics,¹⁰⁵ plasma expanders,¹⁰⁶ and other substances. There is large sample-to-sample variation in the type and composition of proteins present, making accurate measurement difficult. Turbidimetric methods, which are commonly used, are imprecise, with a coefficient of variation as high as 20%.¹⁰⁷ Currently there is no reference measurement procedure or standardized reference material for urinary protein. Each of the different methods in use has differing sensitivity and specificity for the diverse range of proteins found in urine, potentially leading to divergent results. The range of methods and calibrants in use means that between-laboratory variation is unavoidable. Most laboratories currently use turbidimetric or colorimetric measures, which tend to react more strongly with albumin than with globulin and other nonalbumin proteins.¹⁰⁸

METHODS TO MEASURE URINARY ALBUMIN

Urinary albumin can be measured in a number of ways. An antibody binding method, in particular immunoturbidimetry,

Table 23.8 Methods of Proteinuria Measurement

Method	Description	Detection Limit (mg/L)	Protein Types	Causes of Falsely Increased Results	Causes of Falsely Decreased Results
Chemical: Biuret	Copper reagent, measures peptide bonds	50	—	—	Tubular/LMW proteins
Kjeldahl	Precipitated nitrogen				
Turbidimetric (sulfosalicylic acid, trichloroacetic acid)	Addition of precipitant denatures protein; suspension's turbidity is read in a densitometer	50–100	Many, including γ -globulin light chains and albumin More sensitive to albumin than to globulins and nonalbumin proteins	Tolmetin sodium (Tolectin), tolbutamide, antibiotics (penicillin, nafcillin, oxacillin), radiocontrast agents	Tubular/LMW proteins
Dye binding (e.g., Coomassie Brilliant Blue, pyrogallol red)	Indicator changes color in presence of protein	50–100	More sensitive to albumin than to globulins and nonalbumin proteins (pyrogallol red improves this shortcoming)	Pyrogallol red: aminoglycoside, gelatin solutions such as plasma expanders	Tubular/LMW proteins

LMW, Low molecular weight.

Adapted from Cameron JS. The patient with proteinuria and or hematuria. In: Davison A, ed. *Oxford Textbook of Clinical Nephrology*. 3rd ed. Oxford: Oxford University Press; 2005:389–411.

is most commonly used.^{74,108} Because a single protein is being measured, performance of albumin assays tend to be superior to total protein assays, at least at low concentrations of protein.¹⁰⁸ However, the urine of healthy individuals contains a range of albumin molecules. Albumin may be immunoreactive, nonimmunoreactive, fragmented, or biochemically modified¹⁰⁹ and the proportions of these different types of albumin molecules in normal urine are variable.

Albumin fragments may be generated during proteolysis of albumin in renal tubules or plasma and may account for a significant proportion of total urinary albumin. A study in subjects with type 1 diabetes found that 99% of albumin was excreted as fragments <10 kDa. Another study showed that albumin fragments constituted up to 30% of total urinary protein in patients with nephrotic syndrome.¹¹⁰ Intact albumin has at least five antigenic sites.¹¹¹ Routine clinical methods use both polyclonal and monoclonal antibodies, which have different sensitivities for the detection of altered forms of albumin.⁹⁸ Nonimmunoreactive forms of albumin also exist; these are either fragments that do not contain the binding sites for the antibody in use in a particular assay or intact albumin in which the epitopes have undergone conformational change.¹¹²

HPLC detects both immunoreactive and nonimmunoreactive albumin. Higher values for urinary albumin are generally seen in HPLC than in immunologic detection methods. This observation led to a hypothesis that there are clinically significant amounts of nonimmunoreactive albumin in urine. In a study exploring this issue, differences in urinary albumin were detected by four immunoassays and HPLC. However, the higher values seen in HPLC techniques may also represent the detection of nonalbumin macromolecules.¹¹³ Conformational change in albumin molecules may be induced by changes in urinary pH, urea, glucose, and ascorbate concentrations. Bilirubin usually occupies a small proportion of albumin

molecules, but in severe hyperbilirubinemia it may bind to >50% of albumin.⁹⁸

DIFFERENT LABORATORY METHODS TO MEASURE ALBUMIN IN THE URINE

The laboratory methods for measuring urinary albumin are as follows:

Immunoturbidimetric technique: Albumin in a sample of urine reacts with a specific antibody. The turbidity is measured with a spectrophotometer, and the absorbency is proportional to the albumin concentration.¹¹⁴

Double-antibody radioimmunoassay: Albumin in a urine sample competes with a known amount of radiolabeled albumin for fixed binding sites of antialbumin antibodies. Free albumin can be separated from bound albumin by immunoabsorption of the (albumin-bound) antibody. Albumin concentration in the resulting sample of albumin-bound antibody is inversely proportional to its radioactivity, which is measured against a standard curve.¹¹⁵ This is a sensitive assay, but its use is limited by its expense and the need that it is to be performed in a laboratory that can manage radioactive substances.¹⁰⁹

Nephelometry: Albumin in the urine sample reacts with a specific antialbumin antibody, forming light-scattering antigen–antibody complexes that can be measured with a laser nephelometer. The amount of albumin is directly proportional to scatter in the signal.¹¹⁶

Size-exclusion HPLC (SE-HPLC): Chromatographic techniques are used to measure both immunoreactive and nonimmunoreactive albumin. Proteins of different sizes are separated as they pass at different speeds through a column containing size-selective gel.¹¹⁷ SE-HPLC is more sensitive for the detection of albumin than the immune-based methods, but its specificity is limited by an inability

to discriminate between albumin and other proteins of the same size, such as globulins.¹⁰⁹

Although the correlation among results obtained using most of these quantitative assays is very good,⁹⁶ a good correlation indicates only precision, a strong linear relationship, but not accuracy in quantifying all clinically significant proteins present. Results obtained by radioimmunoassay, immunoturbidimetry, nephelometry, and HPLC may vary significantly.¹⁰⁹ Therefore, ideally, the same assay should be used when albuminuria results are compared over time for a given patient. The choice of assay used to measure albuminuria is largely determined by issues of accuracy, cost, and convenience. Currently, there is no standardized procedure for measuring urine albumin and reporting results in standardized units; however, considering the recommendations for using ACRs as the standard measure for urinary protein, a number of professional bodies have moved toward establishing standard laboratory collection, measuring, and reporting procedures.⁹⁸ Standardization for measurement of albumin requires a reference material and reference measurement procedure. Using purified albumin as the reference material would not reflect the various molecular forms that may be present in the urine but may be the most practical approach to standardization. Most routine methods for urinary albumin measurement are currently calibrated against dilutions of CRM 470, a higher-order serum protein reference material with an albumin concentration of 39.7 g/L.^{98,108} Other issues that would need to be addressed to standardize the measurement of urinary albumin include clarification of the molecular forms of albumin in freshly voided urine, the degree of degradation that occurs during storage and freezing, and the appropriate upper limits of normal in different age and sex groups.⁹⁸

TIMED VERSUS RANDOM COLLECTION FOR PROTEINURIA ASSESSMENT

Random “spot” specimens for urinary protein, expressed as a concentration, are often inaccurate for estimation of 24-hour levels because of the impact of patient hydration status on urine concentration. There is also variation in protein excretion, which can occur throughout the day (especially resulting from exercise and posture) and from day to day.¹¹⁸ Methods to improve the accuracy of spot urine testing include corrections for urine creatinine and specific gravity.¹¹⁹

Protein or albumin measurement in 24-hour urine collection is generally considered the gold standard for measuring protein excretion. However, it can also be inaccurate, primarily through inaccurate urine collection. Urine creatinine can be measured to judge the adequacy of the 24-hour collection. If creatinine excretion is similar to that in previous 24-hour samples, the collection is likely to be reasonably accurate. If no other collections are available for comparison, the adequacy of collection can be judged from the expected normal range of creatinine excretion. For hospitalized men aged 20 to 50 years, this range was found to be 18.5 to 25.0 mg/kg of body weight per day, and for women of the same age, 16.5 to 22.4 mg/kg per day. These values declined with age, so that for men aged 50 to 70 years, creatinine excretion was 15.7 to 20.2 mg/kg per day, and for women, 11.8 to 16.1 mg/kg per day.⁶⁵ Factors influencing the daily creatinine excretion

include determinants of muscle mass, such as gender, race, age, and BSA.⁹⁷

TRANSLATING URINE PCR AND ACR VALUES INTO TOTAL DAILY PROTEIN MEASUREMENTS

Urine ACR and PCR are obtained by dividing the urine protein concentrations by the urine creatinine concentration and expressing the result as mg/mmol or mg/g. Both enzymatic and Jaffe assays are used for the measurement of creatinine in urine.¹⁹ The ratio-based tests aim to correct for the effects of urine concentration on protein measurements. Overall, these tests have shown greater accuracy and less intraindividual variability than concentrations measured in random samples^{98,119,120} and are more acceptable to patients than 24-hour protein measurements. Intraindividual variability is further reduced by using a first void rather than a daytime collection specimen to measure ACR. However, there remains substantial day-to-day variability in both PCR and ACR.⁹⁸ A positive result should be followed with a second measurement, ideally in an early morning sample, to confirm the result.^{19,97}

Clinicians commonly use these tests as estimations of the 24-hour protein in milligrams by multiplying the PCR in mg/mmol or ACR in mg/mmol by 10 (given an average daily creatinine excretion of 10 mmol) or using the value as given if measured in mg/g. (Although the PCR or ACR in mg/g should be multiplied by 8.8 to get an exact value in mg/mmol, these are estimates of 24-hour levels only.) Despite the reasonable performance of the ratio-based tests to estimate 24-hour protein measurements, their ability to predict the true 24-hour protein for an individual is limited by two major factors. The first is variability in the total daily creatinine excretion, in and between individuals, which affects the ratio. The second is the fluctuations in protein excretion that occur throughout the day. An understanding of the factors that may make a urine ACR or PCR value inaccurate is important for clinicians using these tests.

CORRELATION BETWEEN RATIOS AND 24-HOUR URINE PROTEIN

Summary tables of studies comparing urine ACR and PCR with timed collections for urinary albumin and protein can be found in the 2012 KDIGO-CKD guideline. There is a relatively high degree of correlation between 24-hour urine protein excretion and PCR in random, single-voided urine samples in healthy controls¹²¹ and in patients with a variety of kidney diseases.^{122–125} The correlation has been shown in studies in patients with glomerulonephritis¹²⁶ and with type 1 diabetes mellitus (DM),¹²⁷ and in renal transplant recipients.^{128,129} In some studies, the correlation between PCR and 24-hour protein measurements was less robust when proteinuria was in the nephrotic range and above.^{124,127}

The urine ACR has been shown to correlate well with 24-hour urinary albumin measurements in a number of studies in people with diabetes.¹³⁰ However, a large study evaluating the correlation between ACR and 24-hour albumin excretion in 1186 subjects with type 1 DM enrolled in the Diabetes Control and Complications Trial and its follow-up study, Epidemiology of Diabetes Interventions and Complications, showed that ACR systematically underestimated albumin excretion, particularly in men.¹³¹ Another study in the diabetic

population showed that ACR increased relative to 24-hour albumin excretion with increasing age.¹³² These studies highlight the fact that variability in creatinine excretion in certain groups alters the ratio. Individuals with lower muscle mass, such as females and elderly patients, would be expected to have higher ACRs than those with higher muscle mass for the same level of urinary albumin excretion.

VARIABILITY IN CREATININE EXCRETION

Under normal circumstances, a steady rate of creatinine excretion occurs throughout the day.¹³³ The accuracy of the ratio-based variables depends on a constant excretion rate. This drawback limits their usefulness in the setting of rapidly changing renal function such as AKI, in which creatinine excretion is reduced, increasing the ratio for the same amount of protein excretion.

Creatinine excretion rises with increasing muscle mass, so ACR and PCR are reduced for a given level of protein excretion in groups with higher muscle mass, such as men, younger people, and certain population ancestry groups.^{134,135} The fact that the average urinary creatinine excretion is 40% to 50% higher in men than in women^{131,132} has led some guidelines to recommend gender-specific cutoff values for ACRs, with lower thresholds for men than women. Commonly used urine ACR thresholds for diabetic nephropathy are 25 mg/g (2.5 mg/mmol) for males and 35 mg/g (3.5 mg/mmol) for females.⁹⁷ However, the 2012 KDIGO guidelines recommend an ACR threshold of 3.0 mg/mmol for both sexes, reflecting that the ACR is an estimation with a variety of other variables, such as age and population ancestry, that are not corrected for.¹⁹

FLUCTUATIONS IN PROTEIN EXCRETION

Factors that may cause transient increases in urinary protein excretion include exercise, urinary tract infections, and upright posture. High-intensity exercise may cause transient proteinuria lasting for 24 to 48 hours in healthy subjects.^{136,137} Patients with CKD or diabetes have been shown to have higher levels of urinary protein or albumin excretion after exercise than control subjects.^{138–140} Some guidelines have recommended screening for urinary tract infection if proteinuria is detected. However, a review of available studies suggested that asymptomatic urinary tract infection was unlikely to cause proteinuria and that screening may be unnecessary.¹⁴¹

Upright posture can cause an increase in urine protein excretion in otherwise healthy young adults.¹⁴² Postural proteinuria is usually diagnosed by detecting proteinuria in a random sample taken while that subject has been upright that is absent in a first morning void specimen. It usually does not exceed 1 g in 24 hours. Kidney histologic examination in patients with postural proteinuria generally yields normal or nonspecific findings,^{142,143} and patients with postural proteinuria have been shown to have an excellent long-term prognosis.¹⁴⁴ An increased urine protein excretion found on a random sample in a young person should prompt a testing of an early morning specimen to exclude postural proteinuria.⁹⁸

Diurnal variation in protein excretion occurs in healthy individuals and patients with CKD. Overnight urinary protein

excretion is lower, and the amount less variable, than in the daytime.^{145,146} Thus timing of urine collection is likely to influence the sensitivity and specificity of screening tests for urine protein or albumin excretion. Samples taken at first void are most likely to accurately quantify 24-hour protein or albumin excretion,^{147,148} and first void specimens are therefore regarded as preferable by a number of guidelines.^{19,72,97}

URINARY ACR VERSUS PCR

Because the relationship between albumin excretion and total protein excretion is nonlinear,¹⁴⁹ an ACR cannot be derived from a PCR, and vice versa. The ACR, rather than PCR, has been recommended by a number of guidelines because of an improved ability to standardize urinary albumin versus total protein measurement and the fact that albumin is the predominant protein lost in the urine.^{19,97} These advantages have already been outlined. ACR has not been shown to be superior to PCR in determining prognosis or detecting CKD in nondiabetic subjects.⁹⁷ A retrospective cohort study comparing urine PCR, ACR, and 24-hour protein at a single center showed that the three were equal in predictive utility for doubling of serum creatinine, commencement of renal replacement therapy, and all-cause mortality.¹⁴⁹

REAGENT STRIP TESTING

Multireagent dipstick urinalysis has been used widely as an initial screening tool for the evaluation of proteinuria because of its low cost, availability, and ability to provide rapid point-of-care information to clinicians. Most dipstick reagents are semiquantitative, containing a pH-sensitive colorimetric indicator that changes color when negatively charged proteins bind to it. Dipstick testing for protein has limited sensitivity for nonalbumin and positively charged proteins^{64,150–152} and therefore often has false-negative results in the presence of predominantly LMW (tubular or overflow) proteinuria.¹⁰⁹ Albumin-specific dipsticks may also be used.

The dipstick tests protein or albumin concentration, rather than an excretion rate, so it is strongly affected by changes in urine concentration. Very dilute urine may give false-negative results, and concentrated urine may give false-positive results. Measuring specific gravity concurrently with urinary protein on a dipstick can thus help with interpretation of a urine dipstick result.¹⁵³ A very high urine pH (>7.0) can give false-positive results, as can contamination of the urine with blood. Operator-dependent differences may also occur with manual reading of dipsticks, decreasing reproducibility.^{147,151} The use of automated reader devices improves interoperator variability.

ALBUMIN-SPECIFIC DIPSTICKS

In addition to dipsticks that are designed to measure total protein, albumin-specific dipsticks are in use. Many of these use dye binding methods^{154–156} but antibody-based detection methods are also available.¹⁵⁷ Several studies have examined the sensitivity and specificity of the newer reagent strips that measure very low concentrations of urine albumin. Most of these investigations studied patients with diabetes, and most examined the Micral-Test^{155,158–161} (Boehringer Mannheim, Mannheim, Germany), the Micro-Bumintest^{155,162} (Ames

Division, Miles Laboratories, Elkhart, Indiana), or both. In general, these albumin reagent strip tests are more sensitive than standard dipstick tests, but they also have a relatively high rate of false-positive results.

NEW DEVICES FOR POINT-OF-CARE PROTEIN TESTING

New systems with a creatinine test pad can report ACRs in the range previously classified as “microalbuminuria” (corresponding to <300 mg/g) or total PCRs. They overcome some of the error inherent in measuring urinary protein concentrations rather than protein excretion rates. The CLINITEK system (CLINITEK microalbumin, Siemens Medical Solutions Diagnostics, Mishawaka, Indiana) uses a reagent strip with two pads, using a dye binding method for albumin measurement and a creatinine assay based on the peroxidase-like activity of a copper creatinine complex. The reagent strips are used in combination with an analyzer to give a semiquantitative assessment of ACR.^{163,164}

ROLE OF POINT-OF-CARE AND REAGENT STRIP TESTING

The role of urinary reagent strip testing, using either protein or albumin-specific dipsticks, in the general and high-risk population is the subject of debate. Most guidelines do not recommend the use of the urine dipstick as an initial screening test for proteinuria.⁷⁴ However, reagent strip testing may be of particular use in settings where laboratory access is limited. Newer point-of-care devices that can measure low levels of albuminuria and provide ACRs may have a role to play in population screening, but this role has not yet been defined by large studies.¹⁹

General Population

Observational studies have shown that reagent strip–proven proteinuria is associated with progression to ESKD and mortality in the general population.^{68,70} However, general population screening could lead to unnecessary investigations, possible harm, and excess costs.^{165–168} As with all diagnostic tests, the positive and negative predictive values of the urinary reagent strips and point-of-care devices depend on the setting in which they are used as well as their sensitivity and specificity.^{169,170} A study comparing urinary ACRs with protein dipsticks (Bayer Multistix) using data from the previously mentioned AusDiab study showed that positive predictive values varied greatly across low- and high-risk subgroups. The dipstick test showed a good ability to rule out proteinuria, with a reagent strip result of less than trace having a negative predictive value of 97.6% for ACR values of 30 mg/g or higher and a negative predictive value of 100% for ACR values of 300 mg/g or higher.¹⁷¹ The investigators concluded that urine reagent strip testing is a reasonable “rule-out” test for proteinuria. However, an analysis by Samal and Linder argues that the rate of false-positive results, which was as high as 53% using an ACR cutoff of 30 mg/g or higher, makes urinary reagent strip testing unacceptable in the general population owing to the cost, anxiety, and workload generated by false-positive results.¹⁶⁸ Because of the high rate of false-positive results, a positive reagent strip test result mandates confirmation with a quantitative test. This is a factor that significantly limits the cost-effectiveness of reagent strip testing for population screening.^{166,168}

Screening of schoolchildren with urinary reagent strips in Japan, Taiwan, and Korea can detect asymptomatic renal disease at an early stage.^{172–174} However, there are no data to show that this policy results in improved outcomes or has benefits from a health economics perspective. Several studies have used models to assess the benefits of general population screening with urinary reagent strips, followed by angiotensin-converting enzyme inhibitor or angiotensin-receptor blocker use in the proteinuric population. One such study, assessing the utility of general practitioner–led general population screening for proteinuria in Australia in 2002, concluded that there was insufficient evidence to support this practice.¹⁶⁶ A study that assessed this practice in a nondiabetic, nonhypertensive U.S. population using cost per quality-adjusted life year concluded that it was cost-effective only if selectively directed toward high-risk groups (aged >60 years or with hypertension) or conducted at long intervals (10 years).¹⁶⁷

High-Risk Populations

In contrast to general population screening, there are several studies showing the cost-effectiveness of high-risk population screening for proteinuria with urinary dipstick testing^{167,175,176} and subsequent antiproteinuric therapy. A model based on screening of a hypertensive, diabetic U.S. population cohort with Micral-II semiquantitative reagent strips for albumin and initiating irbesartan treatment in patients who had microalbuminuria (estimated 24-hour urinary albumin excretion >20 $\mu\text{g}/\text{min}$) or higher levels of urinary albumin showed a 44% reduced incidence in the cumulative incidence of ESKD and was likely to be cost-effective.¹⁷³ Screening high-risk populations, such as patients with diabetes, hypertension, or known vascular disease, for microalbuminuria is recommended in most guidelines,^{19,177,178} although the frequency at which testing should occur is either not specified or inconsistent. Guidelines generally advise the use of laboratory rather than reagent strip testing in the high-risk population.¹⁷⁹ Studies using newer devices such as the CLINITEK system have shown good negative predictive values, making it an effective rule-out test.^{180–182} The CLINITEK system has shown good performance in diabetic, general population,¹⁶³ and CKD cohorts.¹⁸³ However, the usefulness and cost-effectiveness of these newer devices are yet to be confirmed by large studies.

PROTEINURIA MEASUREMENT IN SPECIFIC POPULATIONS

PREGNANT PATIENTS

Proteinuria is generally defined as ≥ 300 mg/24 hours for the diagnosis of preeclampsia. The roughly equivalent urine PCR of 300 mg/g (30 mg/mmol) showed reasonable performance in estimating 24-hour protein excretion and as a rule-out test in two systematic reviews of studies in this setting, although there were no data on PCR for predicting outcomes. Currently, there is insufficient evidence to substitute urine albumin measurement for total protein in pregnant women with hypertension or suspected preeclampsia.^{184,185}

CHILDREN

Normal levels of protein excretion in children are <10 mg/ m^2/day , or <4 mg/ m^2/h .¹⁸⁶ Nephrotic-range proteinuria is defined as 1000 mg/ m^2/day , or 40 mg/ m^2/h , or higher.

Urine PCR has been shown to have reasonable accuracy in reflecting 24-hour levels of protein excretion in a small study of 15 children.¹⁸⁷ PCR has been shown to predict an increased rate of GFR decline in children.^{188–190} There is currently insufficient evidence linking elevated urine ACR with adverse outcomes to recommend the use of ACR in a pediatric population.¹⁹¹ A number of studies have shown contradictory data with regard to the relationship of urinary albumin excretion in healthy children and various conditions, such as obesity, hypertension, fasting glucose, and insulin resistance.¹⁹²

KIDNEY TRANSPLANT RECIPIENTS

Proteinuria predicts allograft loss, cardiovascular risk, and death in kidney transplant recipients.¹⁹³ One study showed that 24-hour urinary albumin measurement was superior to total protein measurement in predicting graft loss¹⁹⁴; another using urine ACR and PCR showed equivalent performances.¹⁹⁵

URINE MICROSCOPY

Examination of the urine by microscopy has been performed systematically at least since the 19th century¹⁹⁶ and remains a vital investigation that is often considered a component of the physical examination of a patient in whom kidney disease is suspected.

PREPARATION AND METHOD

Formed elements in urine degrade rapidly and so urine microscopy is best performed on a fresh urine sample within 2 hours of collection. Urine preservation is not a routinely applied technique but has been performed successfully by some institutions. Traditionally, ethanol is used to preserve uroepithelial cells for cytology but this does not prevent lysis of red and white blood cells.¹⁹⁷ Formaldehyde-based solutions have been used to preserve urine sediment for up to 3 months, and commercial preservatives such as buffered boric acid are also available.

The first urine of the morning specimen has been recommended in the past for urine microscopy because it is acidic and concentrated, but a midstream specimen of the second urine of the morning is favored owing to the lysis of urine particles after prolonged storage of urine in the bladder overnight. Specimens should preferably not be refrigerated because this may cause precipitation of crystals. High urine pH and dilute urine lead to more degradation of formed elements.

A guideline provided by the European Confederation of Laboratory Medicine suggests standardization of preparation of the urine sediment for microscopy.¹⁹⁸ A sample of 5–12 mL of urine should be centrifuged at 400 g or 2000 rpm for 5 minutes, a defined volume of supernatant removed by suction rather than arbitrary decanting, and the pellet resuspended by gentle agitation. A drop of urine should be placed on a slide under a coverslip, and it should be examined ideally with phase-contrast microscopy rather than usual brightfield microscopy, at low power ($\times 160$) then at high power ($\times 400$). Staining or polarized light may also be useful to identify certain substances.

CELLS OF THE URINARY SEDIMENT

ERYTHROCYTES

“Hematuria” is defined as three or more erythrocytes per high-power field. Transient hematuria is common and may be due to strenuous exercise. Persistent hematuria on three repeated urine samples warrants investigation. Studies show great variation in the prevalence of microscopic hematuria from as low as 0.18% to as high as 16.1%.¹⁹⁹ A study of more than a million Israeli military recruits showed an incidence of persistent hematuria of 0.3%.²⁰⁰ This finding was associated with an increased risk of subsequent ESKD (hazard ratio = 19.5) although the absolute risk of ESKD was low, at 34.0 per 100,000 person-years.

Even when the urine appears red the sediment should be examined to determine whether red blood cells are present because a red appearance may be due to other causes, such as hemoglobinuria and myoglobinuria. Macroscopic hematuria is much more likely to be due to malignancy. Isomorphic red blood cells, which look similar to the erythrocytes found in the bloodstream, are thought to be nonglomerular in origin. Dysmorphic red blood cells (Fig. 23.3) are erythrocytes from the glomerular capillary, and their irregular appearance is a consequence of damage due to pH and osmolality changes as the cells travel through the tubule. Glomerular hematuria is defined by some institutions in terms of percentage of dysmorphic red blood cells, but the threshold at which this value is believed to be significant is not standardized. Glomerular hematuria is variously defined as more than 10% to 80% dysmorphic red blood cells or more than 2% to 5% acanthocytes, which are a subtype of dysmorphic red blood cells with protruding blebs.

Automated methods of examining for glomerular or nonglomerular hematuria have been developed in an attempt to overcome the problems with reliability and reproducibility of urine microscopy. These methods function by measuring the mean corpuscular volume (MCV) of the red cells. An MCV smaller or larger than the normal range (50–80 fL) is recorded as dysmorphic. The role of urinary red blood cell MCV in diagnosis was reviewed in a meta-analysis,²⁰¹ which concluded that the diagnostic value of this test is limited and that the urinary MCV test was not reliable in cases of low-grade hematuria because of interfering debris.

Causes of hematuria are listed in Table 23.9. Although anticoagulation often unmasks another underlying etiology for hematuria, over-anticoagulation itself may cause glomerular bleeding, as in the case of the relatively newly recognized condition warfarin-induced nephropathy, in which obstruction of tubules by red blood cell casts may cause AKI.²⁰²

URINE CYTOLOGY

Cytology is usually performed and is recommended as part of the investigation for nonglomerular hematuria. Although it is the realm of urologists, nephrologists may frequently encounter patients with nonglomerular hematuria as part of the investigative process for asymptomatic microscopic hematuria. A number of studies have suggested that the value of routine urine cytology as part of the workup for nonglomerular hematuria is limited if other investigations such as imaging and flexible cystoscopy are performed. For example, in a study of 2778 patients²⁰³ who presented to a

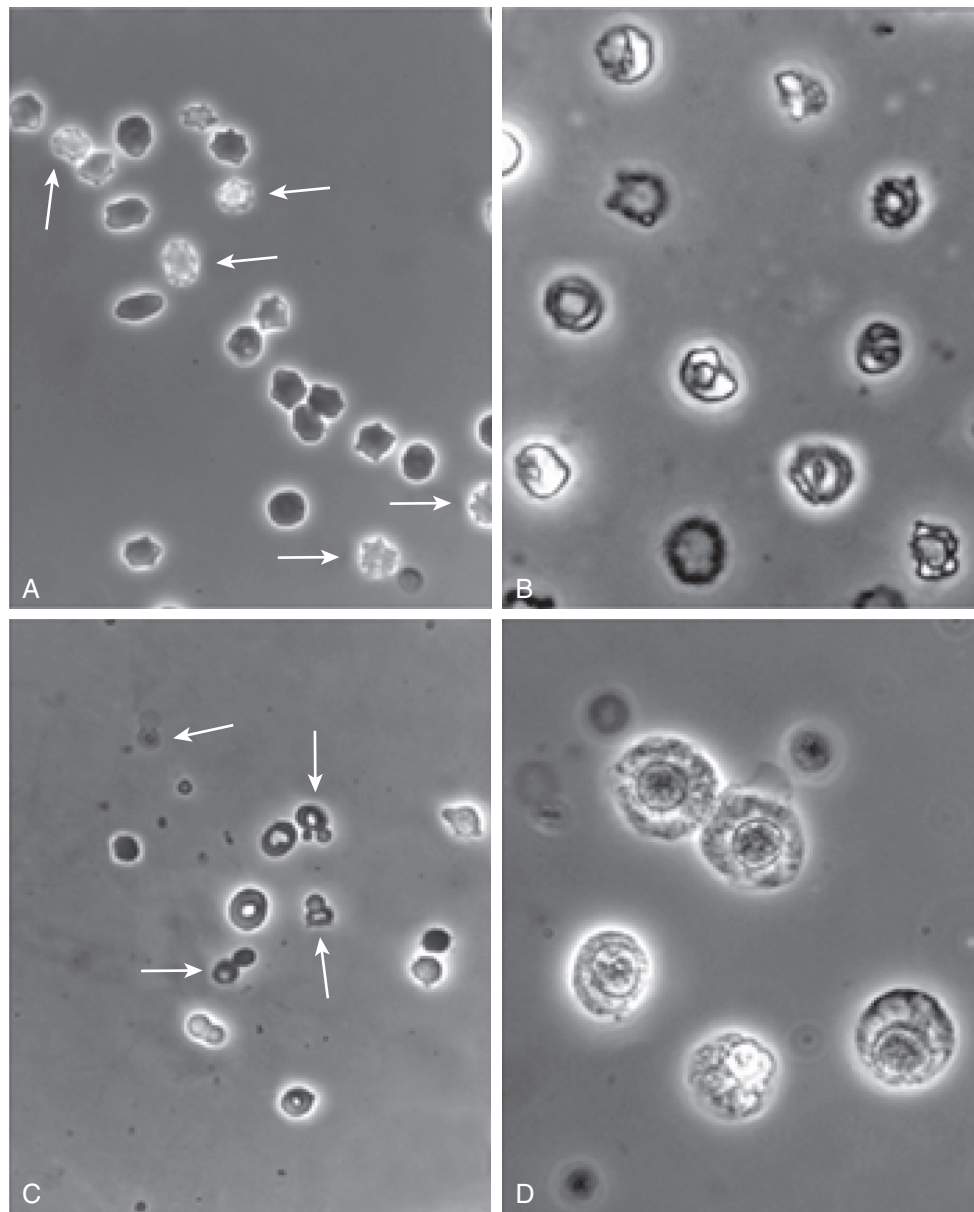


Fig. 23.3 Erythrocytes in urine. (A) Isomorphic erythrocytes, some with a “crenated” appearance (arrows). (B) Different types of dysmorphic erythrocytes. (C) Acanthocytes (arrows). (D) Proximal renal tubule cells. (From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.)

hospital hematuria clinic in the United Kingdom, 974 patients had “nonvisible” or microscopic hematuria. Of the patients with microscopic hematuria, 4.6% had a urothelial malignancy, which in 93% was a bladder tumor. Urothelial cancer cytology demonstrated only 45.5% sensitivity and 89.5% specificity. Only two patients with abnormal urine cytology as the only positive finding had urothelial malignancy on further investigation.

LEUKOCYTES

The most common leukocytes found in the urine, neutrophils, are usually an indication of infection or contamination. Eosinophils are detectable with Wright stain or Hansel stain. Hansel stain has improved sensitivity, especially because a

urine pH of <7 inhibits Wright stain. Although eosinophiluria was initially associated with drug-induced hypersensitivity, the list of diseases that may be associated with eosinophiluria is diverse and includes renal cholesterol embolism, rapidly progressive glomerulonephritis, and prostatitis.

The diagnostic value of the presence of other leukocytes, such as lymphocytes and macrophages, is currently limited, although lymphocytes have been indicative of transplant rejection, and macrophages may be found in glomerulonephritis.

OTHER CELLS

“Squamous cells,” the largest cells of the urinary sediment, derive from the urethra or external genitalia. “Renal tubule

Table 23.9 Causes of Hematuria^a**Glomerular Hematuria****Glomerular Lesions**

Thin basement membrane nephropathy
 Mesangial immunoglobulin A (IgA) glomerulonephritis
 Focal and segmental hyalinosis and sclerosis (focal glomerulosclerosis)
 Lupus glomerulonephritis
 Crescentic glomerulonephritis, including Wegener granulomatosis, microscopic polyangiitis, and Goodpasture syndrome
 Membranous glomerulonephritis
 Mesangiocapillary glomerulonephritis
 Dense deposit disease
 Poststreptococcal glomerulonephritis

Nonglomerular Disease

Autosomal dominant polycystic kidney disease
 Exercise hematuria
 Bleeding diathesis
 Drugs, including anticoagulants

Nonglomerular Hematuria

Urinary tract infection
 Urinary tract calculi
 Hypercalciuria and hyperuricosuria
 Autosomal dominant polycystic kidney disease
 Benign prostatic hypertrophy
 Transitional cell carcinoma
 Renal cell carcinoma
 Prostatic carcinoma
 Exercise hematuria
 Trauma
 Bleeding diathesis and anticoagulants
 Drugs
 Renal papillary necrosis
 Sickle cell disease

^aUnknown whether glomerular or nonglomerular hematuria after percutaneous coronary artery angioplasty.

From Kincaid Smith P, Fairley K. The investigation of haematuria. *Semin Nephrol.* 2005;25:127–135.

epithelial cells” may be present in tubular injury. “Urothelial cells” may be seen in urologic diseases such as malignancy.

OTHER ELEMENTS**LIPIDS**

Lipids appear as spherical, translucent drops of varying size. They may also be within the cytoplasm of cells as “oval fat bodies.” Under polarized light, lipid droplets look like Maltese crosses. Lipiduria is usually associated with heavy proteinuria but may also be present in Fabry disease.

CASTS

“Casts” (Fig. 23.4) are cylindrical bodies of renal origin that form from the aggregation of fibrils of Tamm–Horsfall glycoprotein (uromodulin), which is secreted by cells of the thick ascending limb of the loop of Henle. Trapping of various particles within the cast matrix, as well as degenerative

Table 23.10 Common Casts

Cast	Main Clinical Association(s)
Hyaline	Normal and in renal disease
Granular	Renal disease of any cause
Waxy	Renal disease of any cause
Fatty	Heavy proteinuria
Red blood cell	Proliferative glomerulonephritis, glomerular bleeding
White blood cell	Acute interstitial nephritis, acute pyelonephritis
Tubular epithelial cell	Acute tubular necrosis (“muddy brown” casts), acute interstitial nephritis, proliferative glomerulonephritis

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.

Table 23.11 Common Crystals and Their Appearance

Crystal	Appearance
Uric acid	Usually lozenges but varying shape, yellow-tinged
Calcium oxalate	Monohydrate: ovoid or dumbbell-shaped, biconcave disks Dihydrate: bipyramidal
Calcium phosphate	Prisms, star-like particles or needles of various sizes
Triple phosphate (struvite)	Coffin lids
Cholesterol	Transparent thin plates
Cystine	Hexagonal plates with irregular sides

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.

processes, results in casts with different appearances and clinical significance (Table 23.10). Hyaline casts are non-specific and may be present normally. Granular casts are nonspecific and contain protein aggregates or degenerated cellular elements. Waxy casts are also nonspecific and result from degeneration of other casts. Broad casts are wider waxy casts that are seen in chronic renal failure in which there is dilation of the tubule. Renal tubule epithelial cell casts are formed from the aggregation of desquamated cells of the tubule lining. Because the epithelial cells still appear intact, this finding is usually the result of an acute disease process such as acute tubular necrosis. Red blood cell casts are always pathologic and indicate significant glomerular bleeding, which is often due to rapidly progressive glomerulonephritis.

CRYSTALS

A large variety of crystals can be seen in the urine that may be of diagnostic value (Table 23.11). However, uric acid, calcium oxalate, and calcium phosphate crystals are common and may have little clinical significance because they may precipitate as a result of transient supersaturation of urine

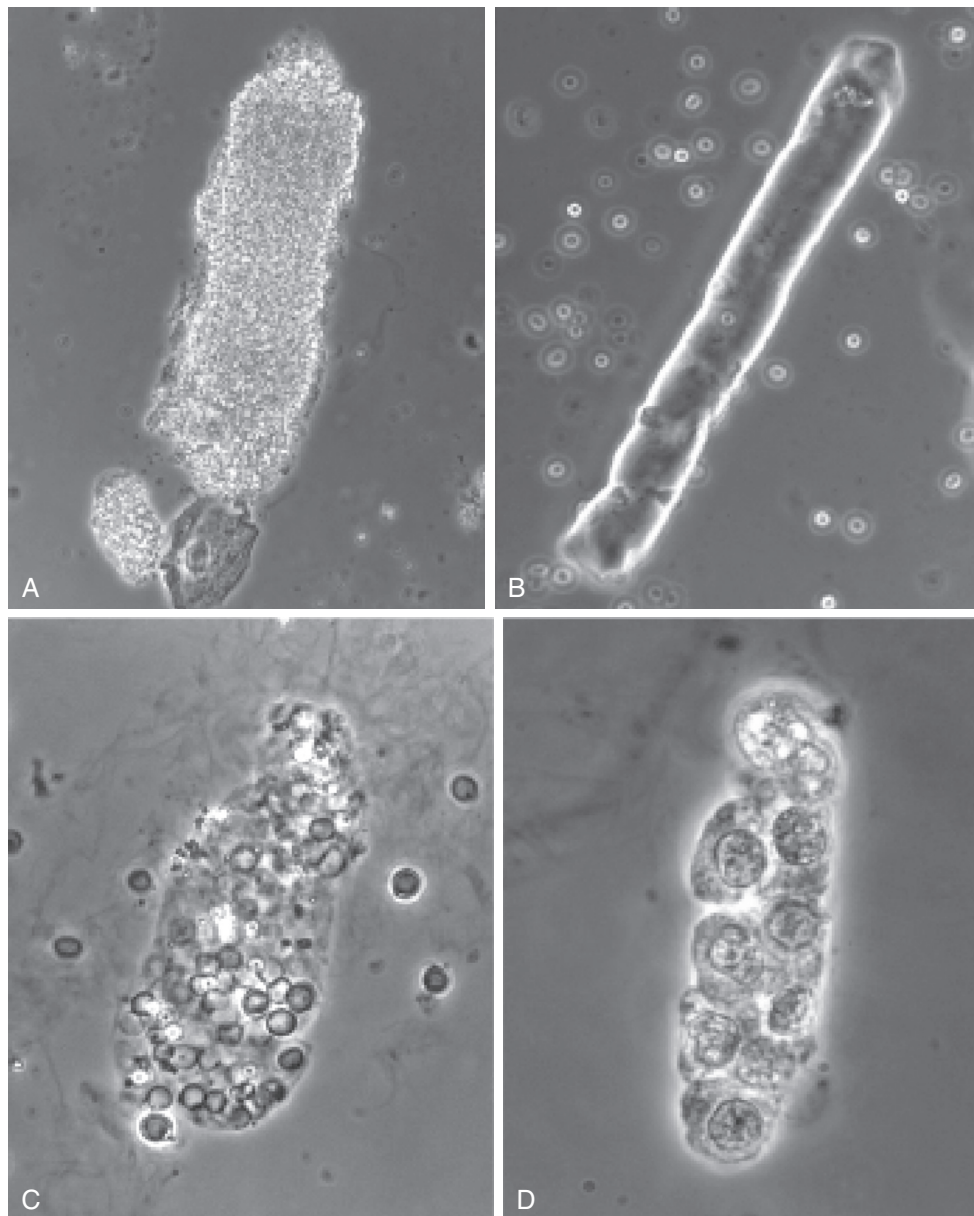


Fig. 23.4 Urinary casts. (A) A finely granular cast. (B) Waxy cast with the typical “melted wax” appearance. (C) A red blood cell cast. (D) A renal tubule epithelial cell cast. (From Stevens LA, Nolin TD, Richardson MM, et al. Comparison of drug dosing recommendations based on measured GFR and kidney function estimating equations. *Am J Kidney Dis.* 2009;54:33–42.)

due to dehydration or cooling of the sample. Of course, depending on the clinical setting, uric acid crystals may be highly significant because they are present in acute uric acid nephropathy as part of tumor lysis syndrome, and calcium oxalate crystals may indicate ethylene glycol poisoning or hyperoxaluria.

A growing list of drugs, beginning with acyclovir and indinavir, may cause crystals in the urine that commonly have unusual shapes.

MICROORGANISMS

Bacteria are commonly seen because of contamination or infection. Fungi such as *Candida*, protozoa such as *Trichomonas*, and parasites such as *Schistosoma* may also be seen.

LIMITATIONS

Urine microscopy depends on expertise and has poor interobserver reliability. In a study involving 10 nephrologists, agreement for various elements in urine microscopy ranged from 31.4% to 79.1% and interobserver agreement was not associated with seniority.²⁰⁴ Despite this limitation, urinary microscopy is still a vital component of the laboratory assessment of renal disease because its findings may crucially influence management of a patient. For example, identification of dysmorphic red blood cells and red blood cell casts in a patient with AKI may prompt empirical treatment with immunosuppressive therapy in advance of results of serology or renal biopsy.

SUMMARY

Despite advances in the field of laboratory assessment of kidney disease, this domain continues to be an evolving science with future research aiming to derive tests and equations that will not only reduce bias but also improve precision and accuracy. There is an ongoing need for better understanding of the intricacies of these tests by physicians and clinical chemists to appropriately diagnose and prognosticate patients and guide their management.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

2. Miller WG, Myers GL, Ashwood ER, et al. Creatinine measurement: state of the art in accuracy and interlaboratory harmonization. *Arch Pathol Lab Med*. 2005;129(3):297–304.
5. Dharnidharka VR, Kwon C, Stevens G. Serum cystatin C is superior to serum creatinine as a marker of kidney function: a meta-analysis. *Am J Kidney Dis*. 2002;40(2):221–226.
19. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.
30. Levey AS, Bosch JP, Lewis JB, et al. A more accurate method to estimate glomerular filtration rate from serum creatinine: a new prediction equation. Modification of Diet in Renal Disease Study Group. *Ann Intern Med*. 2009;130(6):461–470.
31. Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med*. 2009;150:604–612.
32. Matsushita K, Mahmoodi BK, Woodward M, et al. Comparison of risk prediction using the CKD-EPI equation and the MDRD study equation for estimated glomerular filtration rate. *JAMA*. 2012;307(18):1941–1951.
33. Eriksen BO, Mathisen UD, Melsom T, et al. The role of cystatin C in improving GFR estimation in the general population. *Am J Kidney Dis*. 2012;59(1):32–40.
35. Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron*. 2012;16(1):31–41.
37. Schwartz GJ, Haycock GB, Edelmann CM, et al. A simple estimate of glomerular filtration rate in children derived from body length and plasma creatinine. *Pediatrics*. 1976;58(2):259–263.
38. Schwartz GJ, Muñoz A, Schneider MF, et al. New equations to estimate GFR in children with CKD. *J Am Soc Nephrol*. 1976;20(3):629–637.
39. Schwartz GJ, Work DF. Measurement and estimation of GFR in children and adolescents. *Clin J Am Soc Nephrol*. 2009;4(11):1832–1843.
48. Chen S. Retooling the creatinine clearance equation to estimate kinetic GFR when the plasma creatinine is changing acutely. *J Am Soc Nephrol*. 2013;24(6):877–888.
49. Endre ZH, Pickering JW, Walker RJ. Clearance and beyond: the complementary roles of GFR measurement and injury biomarkers in acute kidney injury (AKI). *Am J Physiol Renal Physiol*. 2013;301(4):F697–F707.
52. Stevens LA, Nolin TD, Richardson MM, et al. Comparison of drug dosing recommendations based on measured GFR and kidney function estimating equations. *Am J Kidney Dis*. 2009;54(1):33–42.
55. Kilbride HS, Stevens PE, Eaglestone G, et al. Accuracy of the MDRD (Modification of Diet in Renal Disease) study and CKD-EPI (CKD Epidemiology Collaboration) equations for estimation of GFR in the elderly. *Am J Kidney Dis*. 2013;61(1):57–66.
64. Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis*. 2008;51(6):1052–1067.
65. Israni A, Kasiske B, et al. Laboratory assessment of kidney disease: glomerular filtration rate, urinalysis, and proteinuria. In: Taal M, Chertow G, Marsden P, eds. *Brenner and Rector's the Kidney*. 9th ed. Philadelphia: Saunders; 2011:868–892.
67. Jacobs D, De Mott WR, Willie GR, et al. Urinalysis and clinical microscopy. In: Jacobs D, Kasten BL, De Mott WR, eds. *Laboratory Test Handbook*. Baltimore: Williams & Wilkins; 1990:906–909.
68. Kannel WB, Stampfer MJ, Castelli WP, et al. The prognostic significance of proteinuria: the Framingham study. *Am Heart J*. 1984;108(5):1347–1352.
72. KDOQI. KDOQI clinical practice guidelines and clinical practice recommendations for diabetes and chronic kidney disease. *Am J Kidney Dis*. 2007;49(2 suppl 2):S12–S154.
74. Lamb EJ, MacKenzie F, Stevens PE. How should proteinuria be detected and measured? *Ann Clin Biochem*. 2009;46(Pt 3):205–217.
77. Cameron JS. The patient with proteinuria and or hematuria. In: Davison A, ed. *Oxford Textbook of Clinical Nephrology*. 3rd ed. Oxford: Oxford University Press; 2005:389–411.
78. Brinkkoetter PT, Ising C, Benzing T. The role of the podocyte in albumin filtration. *Nat Rev Nephrol*. 2013;9(6):328–336.
87. Russo LM, Bakris GL, Comper WD. Renal handling of albumin: a critical review of basic concepts and perspective. *Am J Kidney Dis*. 2002;39(5):899–919.
89. Atkins RC, Briganti EM, Zimmet PZ, et al. Association between albuminuria and proteinuria in the general population: the AusDiab study. *Nephrol Dial Transplant*. 2003;18(10):2170–2174.
91. Hillege HL, Fidler V, Diercks GF, et al. Urinary albumin excretion predicts cardiovascular and noncardiovascular mortality in general population. *Circulation*. 2002;106(14):1777–1782.
97. Johnson DW, Jones GR, Mathew TH, et al. Chronic kidney disease and measurement of albuminuria or proteinuria: a position statement. *Med J Aust*. 2012;197(4):224–225.
98. Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion. *Ann Biol Clin (Paris)*. 2010;68(1):9–25.
108. Martin H. Laboratory measurement of urine albumin and urine total protein in screening for proteinuria in chronic kidney disease. *Clin Biochem Rev*. 2011;32(2):97–102.
109. Viswanathan G, Upadhyay A. Assessment of proteinuria. *Adv Chronic Kidney Dis*. 2011;18(4):243–248.
111. Sviridov D, Drake SK, Hortin GL. Reactivity of urinary albumin (microalbumin) assays with fragmented or modified albumin. *Clin Chem*. 2008;54(1):61–68.
118. Naresh CN, Hayen A, Craig JC, et al. Day-to-day variability in spot urine protein:creatinine ratio measurements. *Am J Kidney Dis*. 2012;60(4):561–566.
119. Newman DJ, Pugia MJ, Lott JA, et al. Urinary protein and albumin excretion corrected by creatinine and specific gravity. *Clin Chim Acta*. 2000;294(1–2):139–155.
124. Ruggenenti P, Gaspari F, Perna A, et al. Cross sectional longitudinal study of spot morning urine protein:creatinine ratio, 24 hour urine protein excretion rate, glomerular filtration rate, and end stage renal failure in chronic renal disease in patients without diabetes. *BMJ*. 1998;316(7130):504–509.
125. Price CP, Newall RG, Boyd JC. Use of protein:creatinine ratio measurements on random urine samples for prediction of significant proteinuria: a systematic review. *Clin Chem*. 2005;51(9):1577–1586.
142. Robinson RR. Isolated proteinuria in asymptomatic patients. *Kidney Int*. 1980;18:395–406.
148. Witte EC, Lambers Heerspink HJ, de Zeeuw D, et al. First morning voids are more reliable than spot urine samples to assess microalbuminuria. *J Am Soc Nephrol*. 2009;20(2):436–443.
149. Methven S, Macgregor MS, Traynor JP, et al. Comparison of urinary albumin and urinary total protein as predictors of patient outcomes in CKD. *Am J Kidney Dis*. 2011;57(1):21–28.
165. Iseki K, Kinjo K, Iseki C, et al. Relationship between predicted creatinine clearance and proteinuria and the risk of developing ESRD in Okinawa, Japan. *Am J Kidney Dis*. 2004;44(5):806–814.
166. Craig JC, Barratt A, Cumming R, et al. Feasibility study of the early detection and treatment of renal disease by mass screening. *Intern Med J*. 2002;32(1–2):6–14.
167. Boulware LE, Jaar BG, Tarver-Carr ME, et al. Screening for proteinuria in US adults: a cost-effectiveness analysis. *JAMA*. 2003;290(23):3101–3114.
168. Samal L, Linder JA. The primary care perspective on routine urine dipstick screening to identify patients with albuminuria. *Clin J Am Soc Nephrol*. 2013;8(1):131–135.
186. Hogg RJ, Portman RJ, Milliner D, et al. Evaluation and management of proteinuria and nephrotic syndrome in children: recommendations from a pediatric nephrology panel established at the National Kidney Foundation Conference on proteinuria, albuminuria, risk, assessment, detection, and elimination (PARADE). *Pediatrics*. 2000;105(6):1242–1249.

195. Panek R, Lawen T, Kiberd BA. Screening for proteinuria in kidney transplant recipients. *Nephrol Dial Transplant*. 2011;26(4):1385–1387.
200. Vivante A, Afek A, Frenkel-Nir Y, et al. Persistent asymptomatic isolated microscopic hematuria in Israeli adolescents and young adults and risk for end-stage renal disease. *JAMA*. 2011;306(7):729–736.
201. Offringa M, Benbassat J. The value of urinary red cell shape in the diagnosis of glomerular and post-glomerular haematuria. A meta-analysis. *Postgrad Med J*. 1992;68(802):648–654.
203. Mishriki SF, Aboumarzouk O, Vint R, et al. Routine urine cytology has no role in hematuria investigations. *J Urol*. 2013;189(4):1255–1258.
204. Wald R, Bell CM, Nisenbaum R, et al. Interobserver reliability of urine sediment interpretation. *Clin J Am Soc Nephrol*. 2009;4(3):567–571.

REFERENCES

- Dunn SR, Gabuzda GM, Superdock KR. Induction of bacterial creatininase activity in chronic renal failure: timing of creatinine degradation and effect of antibiotics. *Am J Kidney Dis*. 1997;29(1):72–77.
- Miller WG, Myers GL, Ashwood ER, et al. Creatinine measurement: state of the art in accuracy and interlaboratory harmonization. *Arch Pathol Lab Med*. 2005;129(3):297–304.
- Selvin E, Juraschek SP, Eckfeldt J, et al. Within-person variability in kidney measures. *Am J Kidney Dis*. 2013;61(5):716–722.
- Schwartz GJ, Schneider MF, Maier PS, et al. Improved equations estimating GFR in children with chronic kidney disease using an immunonephelometric determination of cystatin C. *Kidney Int*. 2012;82(4):445–453.
- Dharnidharka VR, Kwon C, Stevens G. Serum cystatin C is superior to serum creatinine as a marker of kidney function: a meta-analysis. *Am J Kidney Dis*. 2002;40(2):221–226.
- Levey AS, Fan L, Eckfeldt JH, et al. Cystatin C for glomerular filtration rate estimation: coming of age. *Clin Chem*. 2014;60:916–919.
- Eckfeldt JH, Karger AB, Miller WG, et al. Performance in measurement of serum cystatin C by laboratories participating in the College of American pathologists 2014 CYS survey. *Arch Pathol Lab Med*. 2015;139:888–893.
- Bargnoux AS, et al. Multicenter evaluation of cystatin C measurement after assay standardization. *Clin Chem*. 2017;63:833–841.
- Stevens LA, Coresh J, Schmid CH, et al. Estimating GFR using serum cystatin C alone and in combination with serum creatinine: a pooled analysis of 3,418 individuals with CKD. *Am J Kidney Dis*. 2008;51:395–406.
- Inker LA, Schmid CH, Tighiouart H, et al. Estimating glomerular filtration rate from serum creatinine and cystatin C. *N Engl J Med*. 2012;367:20–29.
- Schaeffner ES, Ebert N, Delanaye P. Two novel equations to estimate kidney function in persons aged 70 years or older. *Ann Intern Med*. 2012;157:471–481.
- Omuse G, et al. Comparison of equations for estimating glomerular filtration rate in screening for chronic kidney disease in asymptomatic black Africans: a cross sectional study. *BMC Nephrol*. 2017;18:369.
- Pottel H, et al. Estimating glomerular filtration rate for the full age spectrum from serum creatinine and cystatin C. *Nephrol Dial Transplant*. 2017;32:497–507.
- Grubb A, et al. Generation of a new cystatin C-based estimating equation for glomerular filtration rate by use of 7 assays standardized to the international calibrator. *Clin Chem*. 2014;60:974–986.
- Rule AD, Bergstralh EJ, Slezak JM, et al. Glomerular filtration rate estimated by cystatin C among different clinical presentations. *Kidney Int*. 2006;69(2):399.
- White C, Akbari A, Hussain A, et al. Estimating glomerular filtration rate in kidney transplantation: a comparison between serum creatinine and cystatin C based methods. *J Am Soc Nephrol*. 2005;16(12):3763–3770.
- Poge U, Gerhardt T, Stoffel-Wagner B, et al. Calculation of glomerular filtration rate based on cystatin C in cirrhotic patients. *Nephrol Dial Transplant*. 2006;21(3):660. [Epub 2005 Dec2].
- Stevens LA, Padala S, Levey AS. Advances in glomerular filtration rate-estimating equations. *Curr Opin Nephrol Hypertens*. 2010;19(3):298.
- Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.
- Inker LA, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for the evaluation and management of CKD. *Am J Kidney Dis*. 2014;63:713–735.
- White CA, Akbari A, Doucette S, et al. A novel equation to estimate glomerular filtration rate using beta-trace protein. *Clin Chem*. 2007;53(11):1965–1968.
- Pöge U, Gerhardt T, Stoffel-Wagner B, et al. Beta-trace protein-based equations for calculation of GFR in renal transplant recipients. *Am J Transplant*. 2008;8(3):608–615.
- Argyropoulos CP, Chen SS, Ng YH, et al. Rediscovering beta-2 microglobulin as a biomarker across the spectrum of kidney diseases. *Front Med (Lausanne)*. 2017;4:73.
- Inker LA, Tighiouart H, Coresh J, et al. GFR estimation using β -trace protein and β 2-microglobulin in CKD. *Am J Kidney Dis*. 2016;67(1):40–48.
- Spanaus KS, Kollerits B, Ritz E, et al. Serum creatinine, cystatin C, and beta-trace protein in diagnostic staging and predicting progression of primary nondiabetic chronic kidney disease. *Clin Chem*. 2010;56(5):740–749.
- Astor BC, Shafi T, Hoogeveen RC, et al. Novel markers of kidney function as predictors of ESRD, cardiovascular disease, and mortality in the general population. *Am J Kidney Dis*. 2012;59(5):653–662.
- Karger AB, Inker LA, Coresh J, et al. Novel filtration markers for GFR estimation. *J Int Feder Clin Chem Lab Med*. 2017;28(4):277–288.
- Wang X, Lewis J, Appel L, et al. Validation of creatinine-based estimates of GFR when evaluating risk factors in longitudinal studies of kidney disease. *J Am Soc Nephrol*. 2006;17(10):2900–2909.
- Padala S, Tighiouart H, Inker LA, et al. Accuracy of a GFR estimating equation over time in people with a wide range of kidney function. *Am J Kidney Dis*. 2012;60(2):217–224.
- Levey AS, Bosch JP, Lewis JB, et al. A more accurate method to estimate glomerular filtration rate from serum creatinine: a new prediction equation. Modification of Diet in Renal Disease Study Group. *Ann Intern Med*. 1999;130(6):461–470.
- Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med*. 2009;150:604–612.
- Matsushita K, Mahmoodi BK, Woodward M, et al. Comparison of risk prediction using the CKD-EPI equation and the MDRD study equation for estimated glomerular filtration rate. *JAMA*. 2012;307(18):1941–1951.
- Eriksen BO, Mathisen UD, Melsom T, et al. The role of cystatin C in improving GFR estimation in the general population. *Am J Kidney Dis*. 2012;59(1):32–40.
- Go AS, Chertow GM, Fan D, et al. Chronic kidney disease and the risks of death, cardiovascular events and hospitalization. *N Engl J Med*. 2004;351:1296–1305.
- Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron*. 2012;16(1):31–41.
- Stevens LA, Levey AS. Measured GFR as a confirmatory test for estimated GFR. *J Am Soc Nephrol*. 2009;20(11):2305F–2313F.
- Schwartz GJ, Haycock GB, Edelmann CM, et al. A simple estimate of glomerular filtration rate in children derived from body length and plasma creatinine. *Pediatrics*. 1976;58(2):259–263.
- Schwartz GJ, Muñoz A, Schneider MF, et al. New equations to estimate GFR in children with CKD. *J Am Soc Nephrol*. 1976;20(3):629–637.
- Schwartz GJ, Work DF. Measurement and estimation of GFR in children and adolescents. *Clin J Am Soc Nephrol*. 2009;4(11):1832–1843.
- Liu X, Chen J, Wang C. Assessment of glomerular filtration rate in elderly Chinese patients with chronic kidney disease. *Int Urol Nephrol*. 2013.
- Koppe L, Klich A, Ecochard R, et al. Performance of creatinine based equations compared in older patients. *J Nephrol*. 2013;26(4):716–723.
- Maynard SE, Thadhani R. Pregnancy and the kidney. *J Am Soc Nephrol*. 2009;20(1):14–22.
- Dunlop W. Serial changes in hemodynamics during normal human pregnancy. *Br J Obstet Gynaecol*. 1981;88:1–9.
- Park S, Lee SM, Park JS, et al. Midterm eGFR and adverse pregnancy outcomes: the clinical significance of gestational hyperfiltration. *Clin J Am Soc Nephrol*. 2017;12(7):1048–1056.
- Bjornstad P, Cherney D. Kidney function can predict pregnancy outcomes. *CJASN*. 2017.
- Macedo E, Bouchard J, Soroko SH, et al. Fluid accumulation, recognition and staging of acute kidney injury in critically-ill patients. *Crit Care*. 2010;14(3):R82.
- Pickering JW, Ralib AM, Endre ZH. Combining creatinine and volume kinetics identifies missed cases of acute kidney injury following cardiac arrest. *Crit Care*. 2013;17(1):R7.
- Chen S. Retooling the creatinine clearance equation to estimate kinetic GFR when the plasma creatinine is changing acutely. *J Am Soc Nephrol*. 2013;24(6):877–888.
- Endre ZH, Pickering JW, Walker RJ. Clearance and beyond: the complementary roles of GFR measurement and injury biomarkers in acute kidney injury (AKI). *Am J Physiol Renal Physiol*. 2013;301(4):F697–F707.
- U.S. Department of Health and Human Services. Food and Drug Administration. Center for Drug Evaluation and Research (CDER). *Guidance for Industry. Pharmacokinetics in patients with impaired renal function—study design, data analysis and impact on dosing and labeling*. Washington, DC: CDER, 2010.
- Levey AS, Inker LA. Assessment of glomerular filtration rate in health and disease: a state of the art review. *Clin Pharmacol Ther*. 2017;102:405–419. doi:10.1002/cpt.729.

52. Stevens LA, Nolin TD, Richardson MM, et al. Comparison of drug dosing recommendations based on measured GFR and kidney function estimating equations. *Am J Kidney Dis.* 2009;54(1):33–42.
53. Gill J, Malyuk R, Djurdjev O, et al. Use of GFR equations to adjust drug doses in an elderly multi-ethnic group—a cautionary tale. *Nephrol Dial Transplant.* 2007;22(10):2894–2899.
54. Park EJ, Wu K, Mi Z, et al. A systematic comparison of Cockcroft-gault and Modification of Diet in Renal Disease equations for classification of kidney dysfunction and dosage adjustment. *Ann Pharmacother.* 2012;46(9):1174–1187.
55. Kilbride HS, Stevens PE, Eaglestone G, et al. Accuracy of the MDRD (Modification of Diet in Renal Disease) study and CKD-EPI (CKD Epidemiology Collaboration) equations for estimation of GFR in the elderly. *Am J Kidney Dis.* 2013;61(1):57–66.
56. Murata K, Baumann NA, Saenger AK, et al. Relative performance of the MDRD and CKD-EPI equations for estimating glomerular filtration rate among patients with varied clinical presentations. *Clin J Am Soc Nephrol.* 2011;6(8):1963–1972.
57. Pottel H, Hoste L, Dubourg L, et al. As estimated glomerular filtration rate equation for the full age spectrum. *Nephrol Dial Transplant.* 2016;31:798–806.
58. Baylis C, et al. Glucocorticoids and control of glomerular filtration rate. *Semin Nephrol.* 1990;10(4):320–329.
59. vanAcker B, Prummel MF, Weber JA, et al. Effect of prednisone on renal function in man. *Nephron.* 1993;Rannels SR, Jefferson LS. Effects of glucocorticoids on muscle protein turnover in perfused rat hemi-corpus. *Am J Physiol* 1980; 238: E564±72.
60. Beaufrere FJ, Horber FF, Schenk WF, et al. Glucocorticosteroids increase leucine oxidation and impair leucine balance in humans. *Am J Physiol.* 1989;257:E712–E721.
61. Horber FF, Scheidegger J, Frey FJ. Overestimation of renal function in glucocorticosteroid treated patients. *Eur J Clin Pharmacol.* 1985;28:537–541.
62. Santos J, Martins LS. Estimating glomerular filtrations rate in kidney transplantation: still searching for the best marker. *World J Nephrol.* 2015;4(3):345–353.
63. Shaffi SK, Uhlig K, Perrone RD, et al. Performance of creatinine based GFR estimating equations in solid organ transplant recipients. *Am J Kidney Dis.* 2014;63(6):1007–1018.
64. Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51(6):1052–1067.
65. Israni A, Kaisike B. *Laboratory Assessment of Kidney Disease: Glomerular Filtration Rate, Urinalysis, and Proteinuria.* 9th ed. Philadelphia: Saunders; 2011.
66. Rose BD, Post TW. *Clinical Physiology of Acid Base and Electrolyte Disorders.* 5th ed. New York: McGraw Hill; 2001.
67. Jacobs D, De Mott WR, Willie GR, et al. Urinalysis and clinical microscopy. In: Jacobs D, Kasten BL, De Mott WR, eds. *Laboratory Test Handbook.* Baltimore: Williams & Wilkins; 1990:906–909.
68. Kannel WB, Stampfer MJ, Castelli WP, et al. The prognostic significance of proteinuria: the Framingham study. *Am Heart J.* 1984;108(5):1347–1352.
69. Ninomiya T, Perkovic V, de Galan BE, et al. Albuminuria and kidney function independently predict cardiovascular and renal outcomes in diabetes. *J Am Soc Nephrol.* 2009;20(8):1813–1821.
70. Wen CP, Cheng TY, Tsai MK, et al. All-cause mortality attributable to chronic kidney disease: a prospective cohort study based on 462 293 adults in Taiwan. *Lancet.* 2008;371(9631):2173–2182.
71. Jafar TH, Stark PC, Schmid CH, et al. Proteinuria as a modifiable risk factor for the progression of non-diabetic renal disease. *Kidney Int.* 2001;60(3):1131–1140.
72. KDOQI. KDOQI clinical practice guidelines and clinical practice recommendations for diabetes and chronic kidney disease. *Am J Kidney Dis.* 2007;49(suppl 2):S12–S154.
73. Remuzzi G, Benigni A, Remuzzi A. Mechanisms of progression and regression of renal lesions of chronic nephropathies and diabetes. *J Clin Invest.* 2006;116(2):288–296.
74. Lamb EJ, MacKenzie F, Stevens PE. How should proteinuria be detected and measured? *Ann Clin Biochem.* 2009;46(Pt 3):205–217.
75. Hausmann R, Kuppe C, Egger H, et al. Electrical forces determine glomerular permeability. *J Am Soc Nephrol.* 2010;21(12):2053–2058.
76. Haraldsson B, Nyström J, Deen WM. Properties of the glomerular barrier and mechanisms of proteinuria. *Physiol Rev.* 2008;88(2):451–487.
77. Cameron JS. The patient with proteinuria and or hematuria. In: Davison A, ed. *Oxford Textbook of Clinical Nephrology.* 3rd ed. Oxford: Oxford University Press; 2005:389–411.
78. Brinkkoetter PT, Ising C, Benzing T. The role of the podocyte in albumin filtration. *Nat Rev Nephrol.* 2013;9(6):328–336, 46.
79. Huber TB, Benzing T. The slit diaphragm: a signaling platform to regulate podocyte function. *Curr Opin Nephrol Hypertens.* 2005; 14(3):211–216.
80. Löwik MM, Groenen PJ, Levchenko EN, et al. Molecular genetic analysis of podocyte genes in focal segmental glomerulosclerosis—a review. *Eur J Pediatr.* 2009;168(11):1291–1304.
81. Benoit G, Machuca E, Antignac C. Hereditary nephrotic syndrome: a systematic approach for genetic testing and a review of associated podocyte gene mutations. *Pediatr Nephrol.* 2010;25(9): 1621–1632.
82. Tryggvason K, Patrakka J, Wartiovaara J. Hereditary proteinuria syndromes and mechanisms of proteinuria. *N Engl J Med.* 2006;354(13):1387–1401.
83. Comper WD, Glasgow EF. Charge selectivity in kidney ultrafiltration. *Kidney Int.* 1995;47(5):1242–1251.
84. Bolton GR, Deen WM, Daniels BS. Assessment of the charge selectivity of glomerular basement membrane using ficoll sulfate. *Am J Physiol.* 1998;274(5 Pt 2):F889–F896.
85. Burne MJ, Adal Y, Cohen N, et al. Anomalous decrease in dextran sulfate clearance in the diabetic rat kidney. *Am J Physiol.* 1998;274 (4 Pt 2):F700–F708.
86. Asgeirsson D, Venturoli D, Rippe B, et al. Increased glomerular permeability to negatively charged ficoll relative to neutral ficoll in rats. *Am J Physiol Renal Physiol.* 2006;291(5):F1083–F1089.
87. Russo LM, Bakris GL, Comper WD. Renal handling of albumin: a critical review of basic concepts and perspective. *Am J Kidney Dis.* 2002;39(5):899–919.
88. Christov M, Alper SL. Tubular transport: core curriculum 2010. *Am J Kidney Dis.* 2010;56(6):1202–1217.
89. Atkins RC, Briganti EM, Zimmet PZ, et al. Association between albuminuria and proteinuria in the general population: the AusDiab study. *Nephrol Dial Transplant.* 2003;18(10):2170–2174.
90. Levey AS, de Jong PE, Coresh J, et al. The definition, classification, and prognosis of chronic kidney disease: a KDIGO controversies Conference report. *Kidney Int.* 2011;80(1):17–28.
91. Hillege HL, Fidler V, Diercks GF, et al. Urinary albumin excretion predicts cardiovascular and noncardiovascular mortality in general population. *Circulation.* 2002;106(14):1777–1782.
92. Obermayr RP, Temml C, Knechtelsdorfer M, et al. Predictors of new-onset decline in kidney function in a general middle-European population. *Nephrol Dial Transplant.* 2008;23(4):1265–1273.
93. Kronborg J, Solbu M, Njølstad I, et al. Predictors of change in estimated GFR: a population-based 7-year follow-up from the Tromsø study. *Nephrol Dial Transplant.* 2008;23(9):2818–2826.
94. Astor BC, Matsushita K, Gansevoort RT, et al. Lower estimated glomerular filtration rate and higher albuminuria are associated with mortality and end-stage renal disease. A collaborative meta-analysis of kidney disease population cohorts. *Kidney Int.* 2011;79(12):1331–1340.
95. Newman DJ, Thakkar H, Medcalf EA, et al. Use of urine albumin measurement as a replacement for total protein. *Clin Nephrol.* 1995;43(2):104–109.
96. Ballantyne FC, Gibbons J, O'Reilly DS. Urine albumin should replace total protein for the assessment of glomerular proteinuria. *Ann Clin Biochem.* 1993;30(Pt 1):101–103.
97. Johnson DW, Jones GR, Mathew TH, et al. Chronic kidney disease and measurement of albuminuria or proteinuria: a position statement. *Med J Aust.* 2012;197(4):224–225.
98. Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion. *Ann Biol Clin (Paris).* 2010;68(1):9–25.
99. Goren MP, Li JT. The Coomassie Brilliant Blue method underestimates drug-induced tubular proteinuria. *Clin Chem.* 1986;32(2): 386–388.
100. Gosling P. Proteinuria. In: Marshall WJ, Bangert SK, eds. *Clinical Biochemistry: Metabolic and Clinical Aspects.* 2nd ed. New York: Churchill Livingstone; 2008:156–174.
101. Wrong OM, Norden AG, Feest TG. Dent's disease: a familial proximal renal tubular syndrome with low-molecular-weight proteinuria, hypercalciuria, nephrocalcinosis, metabolic bone disease, progressive renal failure and a marked male predominance. *QJM.* 1994;87(8):473–493.
102. McTaggart MP, Stevens PE, Price CP. Investigation of apparent non-albuminuric proteinuria in a primary care population. *Clin Chem Lab Med.* 1961;51(10):2013.

103. Pesce AJ, Boreisha I, Pollak VE. Rapid differentiation of glomerular and tubular proteinuria by sodium dodecyl sulfate polyacrylamide gel electrophoresis. *Clin Chim Acta*. 1972;40(1):27–34.
104. Smith ER, Cai MM, McMahon LP, et al. The value of simultaneous measurements of urinary albumin and total protein in proteinuric patients. *Nephrol Dial Transplant*. 2012;27(4):1534–1541.
105. Marshall T, Williams KM. Extent of aminoglycoside interference in the pyrogallol red–molybdate protein assay depends on the concentration of sodium oxalate in the dye reagent. *Clin Chem*. 2004;50(5):934–935.
106. de Keijzer MH, Klasen IS, Branten AJ, et al. Infusion of plasma expanders may lead to unexpected results in urinary protein assays. *Scand J Clin Lab Invest*. 1999;59(2):133–137.
107. McIntyre NJ, Taal MW. How to measure proteinuria? *Curr Opin Nephrol Hypertens*. 2008;17(6):600–603.
108. Martin H. Laboratory measurement of urine albumin and urine total protein in screening for proteinuria in chronic kidney disease. *Clin Biochem Rev*. 2011;32(2):97–102.
109. Viswanathan G, Upadhyay A. Assessment of proteinuria. *Adv Chronic Kidney Dis*. 2011;18(4):243–248.
110. Candiano G, Musante L, Bruschi M, et al. Repetitive fragmentation products of albumin and alpha1-antitrypsin in glomerular diseases associated with nephrotic syndrome. *J Am Soc Nephrol*. 2006;17(11):3139–3148.
111. Sviridov D, Drake SK, Hortin GL. Reactivity of urinary albumin (microalbumin) assays with fragmented or modified albumin. *Clin Chem*. 2008;54(1):61–68.
112. Osicka TM, Houlihan CA, Chan JG, et al. Albuminuria in patients with type 1 diabetes is directly linked to changes in the lysosome-mediated degradation of albumin during renal passage. *Diabetes*. 2000;49(9):1579–1584.
113. Sviridov D, Meilinger B, Drake SK, et al. Coelution of other proteins with albumin during size-exclusion HPLC: implications for analysis of urinary albumin. *Clin Chem*. 2006;52(3):389–397.
114. Harmoinen A, Vuorinen P, Jokela H. Turbidimetric measurement of microalbuminuria. *Clin Chim Acta*. 1987;166(1):85–89.
115. Rowe DJ, Dawney A, Watts GF. Microalbuminuria in diabetes mellitus: review and recommendations for the measurement of albumin in urine. *Ann Clin Biochem*. 1990;27(Pt 4):297–312.
116. Stamp RJ. Measurement of albumin in urine by end-point immunonephelometry. *Ann Clin Biochem*. 1988;25(Pt 4):442–443.
117. Comper WD, Osicka TM. Detection of urinary albumin. *Adv Chronic Kidney Dis*. 2005;12(2):170–176.
118. Naresh CN, Hayen A, Craig JC, et al. Day-to-day variability in spot urine protein:creatinine ratio measurements. *Am J Kidney Dis*. 2012;60(4):561–566.
119. Newman DJ, Pugia MJ, Lott JA, et al. Urinary protein and albumin excretion corrected by creatinine and specific gravity. *Clin Chim Acta*. 2000;294(1–2):139–155.
120. Bakker AJ. Detection of microalbuminuria. Receiver operating characteristic curve analysis favors albumin-to-creatinine ratio over albumin concentration. *Diabetes Care*. 1999;22(2):307–313.
121. Lemann J, Doumas BT. Proteinuria in health and disease assessed by measuring the urinary protein/creatinine ratio. *Clin Chem*. 1987;33(2 Pt 1):297–299.
122. Schwab SJ, Christensen RL, Dougherty K, et al. Quantitation of proteinuria by the use of protein-to-creatinine ratios in single urine samples. *Arch Intern Med*. 1987;147(5):943–944.
123. Ginsberg JM, Chang BS, Matarese RA, et al. Use of single voided urine samples to estimate quantitative proteinuria. *N Engl J Med*. 1983;309(25):1543–1546.
124. Ruggenenti P, Gaspari F, Perna A, et al. Cross sectional longitudinal study of spot morning urine protein:creatinine ratio, 24 hour urine protein excretion rate, glomerular filtration rate, and end stage renal failure in chronic renal disease in patients without diabetes. *BMJ*. 1998;316(7130):504–509.
125. Price CP, Newall RG, Boyd JC. Use of protein:creatinine ratio measurements on random urine samples for prediction of significant proteinuria: a systematic review. *Clin Chem*. 2005;51(9):1577–1586.
126. Antunes VV, Veronese FJ, Morales JV. Diagnostic accuracy of the protein/creatinine ratio in urine samples to estimate 24-h proteinuria in patients with primary glomerulopathies: a longitudinal study. *Nephrol Dial Transplant*. 2008;23(7):2242–2246.
127. Rodby RA, Rohde RD, Sharon Z, et al. The urine protein to creatinine ratio as a predictor of 24-hour urine protein excretion in type 1 diabetic patients with nephropathy. The Collaborative Study Group. *Am J Kidney Dis*. 1995;26(6):904–909.
128. Dyson EH, Will EJ, Davison AM, et al. Use of the urinary protein creatinine index to assess proteinuria in renal transplant patients. *Nephrol Dial Transplant*. 1992;7(5):450–452.
129. Steinhäuslin F, Wauters JP. Quantitation of proteinuria in kidney transplant patients: accuracy of the urinary protein/creatinine ratio. *Clin Nephrol*. 1995;43(2):110–115.
130. Zelmanovitz T, Gross JL, Oliveira JR, et al. The receiver operating characteristics curve in the evaluation of a random urine specimen as a screening test for diabetic nephropathy. *Diabetes Care*. 1997;20(4):516–519.
131. Younes N, Cleary PA, Steffes MW, et al. Comparison of urinary albumin-creatinine ratio and albumin excretion rate in the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications Study. *Clin J Am Soc Nephrol*. 2010;5(7):1235–1242.
132. Houlihan CA, Tsalamandris C, Akdeniz A, et al. Albumin to creatinine ratio: a screening test with limitations. *Am J Kidney Dis*. 2002;39(6):1183–1189.
133. Vestergaard P, Leverett R. Constancy of urinary creatinine excretion. *J Lab Clin Med*. 1958;51(2):211–218.
134. Mattix HJ, Hsu CY, Shaykevich S, et al. Use of the albumin/creatinine ratio to detect microalbuminuria: implications of sex and race. *J Am Soc Nephrol*. 2002;13(4):1034–1039.
135. Methven S, MacGregor MS, Traynor JP, et al. Assessing proteinuria in chronic kidney disease: protein-creatinine ratio versus albumin-creatinine ratio. *Nephrol Dial Transplant*. 2010;25(9):2991–2996.
136. Heathcote KL, Wilson MP, Quest DW, et al. Prevalence and duration of exercise induced albuminuria in healthy people. *Clin Invest Med*. 2009;32(4):E261–E265.
137. Poortmans JR. Postexercise proteinuria in humans. Facts and mechanisms. *JAMA*. 1985;253(2):236–240.
138. Mogensen CE, Vittinghus E, Sølling K. Abnormal albumin excretion after two provocative renal tests in diabetes: physical exercise and lysine injection. *Kidney Int*. 1979;16(3):385–393.
139. Vittinghus E, Mogensen CE. Albumin excretion during physical exercise in diabetes. Studies on the effect of insulin treatment and of the renal haemodynamic response. *Acta Endocrinol Suppl (Copenh)*. 1981;242:61–62.
140. Koh KH, Dayanath B, Doery JC, et al. Effect of exercise on albuminuria in people with diabetes. *Nephrology (Carlton)*. 2011;16(8):704–709.
141. Carter JL, Tomson CR, Stevens PE, et al. Does urinary tract infection cause proteinuria or microalbuminuria? A systematic review. *Nephrol Dial Transplant*. 2006;21(11):3031–3037.
142. Robinson RR. Isolated proteinuria in asymptomatic patients. *Kidney Int*. 1980;18:395–406.
143. von Bonsdorff M, Koskenvuo K, Salmi HA, et al. Prevalence and causes of proteinuria in 20-year-old Finnish men. *Scand J Urol Nephrol*. 1981;15(3):285–290.
144. Springberg PD, Garrett LE, Thompson AL, et al. Fixed and reproducible orthostatic proteinuria: results of a 20-year follow-up study. *Ann Intern Med*. 1982;97(4):516–519.
145. Houser MT. Characterization of recumbent, ambulatory, and postexercise proteinuria in the adolescent. *Pediatr Res*. 1987;21(5):442–446.
146. Chachati A, von Frenczell R, Foidart-Willems J, et al. Variability of albumin excretion in insulin-dependent diabetics. *Diabet Med*. 1987;4(5):441–445.
147. Koopman MG, Krediet RT, Koomen GC, et al. Circadian rhythm of proteinuria: consequences of the use of urinary protein:creatinine ratios. *Nephrol Dial Transplant*. 1989;4(1):9–14.
148. Witte EC, Lambers Heerspink HJ, de Zeeuw D, et al. First morning voids are more reliable than spot urine samples to assess microalbuminuria. *J Am Soc Nephrol*. 2009;20(2):436–443.
149. Methven S, MacGregor MS, Traynor JP, et al. Comparison of urinary albumin and urinary total protein as predictors of patient outcomes in CKD. *Am J Kidney Dis*. 2011;57(1):21–28.
150. Gyure WL. Comparison of several methods for semiquantitative determination of urinary protein. *Clin Chem*. 1977;23(5):876–879.
151. Rumley A. Urine dipstick testing: comparison of results obtained by visual reading and with the Bayer CLINITEK 50. *Ann Clin Biochem*. 2000;37(Pt 2):220–221.
152. James GP, Bee DE, Fuller JB. Proteinuria: accuracy and precision of laboratory diagnosis by dip-stick analysis. *Clin Chem*. 1978;24(11):1934–1939.

153. Constantiner M, Sehgal AR, Humbert L, et al. A dipstick protein and specific gravity algorithm accurately predicts pathological proteinuria. *Am J Kidney Dis*. 2005;45(5):833–841.
154. Comper WD, Jerums G, Osicka TM. Deficiency in the detection of microalbuminuria by urinary dipstick in diabetic patients. *Diabetes Care*. 2003;26(11):3195–3196.
155. Tiu SC, Lee SS, Cheng MW. Comparison of six commercial techniques in the measurement of microalbuminuria in diabetic patients. *Diabetes Care*. 1993;16(4):616–620.
156. Pugia MJ, Lott JA, Profitt JA, et al. High-sensitivity dye binding assay for albumin in urine. *J Clin Lab Anal*. 1999;13(4):180–187.
157. Sawicki PT, Heinemann L, Berger M. Comparison of methods for determination of microalbuminuria in diabetic patients. *Diabet Med*. 1989;6(5):412–415.
158. Marshall SM. Screening for microalbuminuria: which measurement? *Diabet Med*. 1991;8(8):706–711.
159. Bangstad HJ, Try K, Dahl-Jørgensen K, et al. New semiquantitative dipstick test for microalbuminuria. *Diabetes Care*. 1991;14(11):1094–1097.
160. Minetti EE, Cozzi MG, Granata S, et al. Accuracy of the urinary albumin titrator stick 'Mical-Test' in kidney-disease patients. *Nephrol Dial Transplant*. 1997;12(1):78–80.
161. Poulsen PL, Hansen B, Amby T, et al. Evaluation of a dipstick test for microalbuminuria in three different clinical settings, including the correlation with urinary albumin excretion rate. *Diabetes Metab*. 1992;18(5):395–400.
162. Tai J, Tze WJ. Evaluation of Micro-Bumintest reagent tablets for screening of microalbuminuria. *Diabetes Res Clin Pract*. 1990;9(2):137–142.
163. Graziani MS, Gambaro G, Mantovani L, et al. Diagnostic accuracy of a reagent strip for assessing urinary albumin excretion in the general population. *Nephrol Dial Transplant*. 2009;24(5):1490–1494.
164. Parsons M, Newman DJ, Pugia M, et al. Performance of a reagent strip device for quantitation of the urine albumin:creatinine ratio in a point of care setting. *Clin Nephrol*. 1999;51(4):220–227.
165. Iseki K, Kinjo K, Iseki C, et al. Relationship between predicted creatinine clearance and proteinuria and the risk of developing ESRD in Okinawa, Japan. *Am J Kidney Dis*. 2004;44(5):806–814.
166. Craig JC, Barratt A, Cumming R, et al. Feasibility study of the early detection and treatment of renal disease by mass screening. *Intern Med J*. 2002;32(1–2):6–14.
167. Boulware LE, Jaar BG, Tarver-Carr ME, et al. Screening for proteinuria in US adults: a cost-effectiveness analysis. *JAMA*. 2003;290(23):3101–3114.
168. Samal L, Linder JA. The primary care perspective on routine urine dipstick screening to identify patients with albuminuria. *Clin J Am Soc Nephrol*. 2013;8(1):131–135.
169. Lachs MS, Nachamkin I, Edelstein PH, et al. Spectrum bias in the evaluation of diagnostic tests: lessons from the rapid dipstick test for urinary tract infection. *Ann Intern Med*. 1992;117(2):135–140.
170. Mulherin SA, Miller WC. Spectrum bias or spectrum effect? Subgroup variation in diagnostic test evaluation. *Ann Intern Med*. 2002;137(7):598–602.
171. White SL, Yu R, Craig JC, et al. Diagnostic accuracy of urine dipsticks for detection of albuminuria in the general community. *Am J Kidney Dis*. 2011;58(1):19–28.
172. Cho BS, Kim SD. School urinalysis screening in Korea. *Nephrology (Carlton)*. 2007;12(suppl 3):S3–S7.
173. Yap HK, Quek CM, Shen Q, et al. Role of urinary screening programmes in children in the prevention of chronic kidney disease. *Ann Acad Med Singapore*. 2005;34(1):3–7.
174. Kitagawa T. Lessons learned from the Japanese nephritis screening study. *Pediatr Nephrol*. 1988;2(2):256–263.
175. Palmer AJ, Valentine WJ, Chen R, et al. A health economic analysis of screening and optimal treatment of nephropathy in patients with type 2 diabetes and hypertension in the USA. *Nephrol Dial Transplant*. 2008;23(4):1216–1223.
176. Johnson DW. Evidence-based guide to slowing the progression of early renal insufficiency. *Intern Med J*. 2004;34(1–2):50–57.
177. Bennett PH, Haffner S, Kasiske BL, et al. Screening and management of microalbuminuria in patients with diabetes mellitus: recommendations to the Scientific Advisory Board of the National Kidney Foundation from an ad hoc committee of the Council on Diabetes Mellitus of the National Kidney Foundation. *Am J Kidney Dis*. 1995;25(1):107–112.
178. Levin A, Hemmelgarn B, Culeton B, et al. Guidelines for the management of chronic kidney disease. *CMAJ*. 2008;179(11):1154–1162.
179. Johnson DW. Global proteinuria guidelines: are we nearly there yet? *Clin Biochem Rev*. 2011;32(2):89–95.
180. Le Floch JP, Marre M, Rodier M, et al. Interest of CLINITEK microalbumin in screening for microalbuminuria: results of a multicentre study in 302 diabetic patients. *Diabetes Metab*. 2001;27(1):36–39.
181. Croal BL, Mutch WJ, Clark BM, et al. The clinical application of a urine albumin:creatinine ratio point-of-care device. *Clin Chim Acta*. 2001;307(1–2):15–21.
182. Meinhardt U, Ammann RA, Flück C, et al. Microalbuminuria in diabetes mellitus: efficacy of a new screening method in comparison with timed overnight urine collection. *J Diabetes Complications*. 2003;17(5):254–257.
183. Guy M, Newall R, Borzomato J, et al. Diagnostic accuracy of the urinary albumin:creatinine ratio determined by the CLINITEK microalbumin and DCA 2000+ for the rule-out of albuminuria in chronic kidney disease. *Clin Chim Acta*. 2009;399(1–2):54–58.
184. Côté AM, Brown MA, Lam E, et al. Diagnostic accuracy of urinary spot protein:creatinine ratio for proteinuria in hypertensive pregnant women: systematic review. *BMJ*. 2008;336(7651):1003–1006.
185. Morris RK, Riley RD, Doug M, et al. Diagnostic accuracy of spot urinary protein and albumin to creatinine ratios for detection of significant proteinuria or adverse pregnancy outcome in patients with suspected pre-eclampsia: systematic review and meta-analysis. *BMJ*. 2012;345:e4342.
186. Hogg RJ, Portman RJ, Milliner D, et al. Evaluation and management of proteinuria and nephrotic syndrome in children: recommendations from a pediatric nephrology panel established at the National Kidney Foundation Conference on proteinuria, albuminuria, risk, assessment, detection, and elimination (PARADE). *Pediatrics*. 2000;105(6):1242–1249.
187. Houser M. Assessment of proteinuria using random urine samples. *J Pediatr*. 1984;104(6):845–848.
188. Wingen AM, Fabian-Bach C, Schaefer F, et al. Randomised multicentre study of a low-protein diet on the progression of chronic renal failure in children. European Study Group of Nutritional Treatment of Chronic Renal Failure in Childhood. *Lancet*. 1997;349(9059):1117–1123.
189. Ardissino G, Testa S, Daccò V, et al. Proteinuria as a predictor of disease progression in children with hypodysplastic nephropathy. Data from the Italkid Project. *Pediatr Nephrol*. 2004;19(2):172–177.
190. Wong CS, Pierce CB, Cole SR, et al. Association of proteinuria with race, cause of chronic kidney disease, and glomerular filtration rate in the chronic kidney disease in children study. *Clin J Am Soc Nephrol*. 2009;4(4):812–819.
191. Tsioufis C, Mazaraki A, Dimitriadis K, et al. Microalbuminuria in the paediatric age: current knowledge and emerging questions. *Acta Paediatr*. 2011;100(9):1180–1184.
192. Flynn JT. Microalbuminuria in children with primary hypertension. *J Clin Hypertens*. 2016;18(10):962–965.
193. Knoll GA. Proteinuria in kidney transplant recipients: prevalence, prognosis, and evidence-based management. *Am J Kidney Dis*. 2009;54(6):1131–1144.
194. Nauta FL, Bakker SJ, van Oeveren W, et al. Albuminuria, proteinuria, and novel urine biomarkers as predictors of long-term allograft outcomes in kidney transplant recipients. *Am J Kidney Dis*. 2011;57(5):733–743.
195. Panek R, Lawen T, Kiberd BA. Screening for proteinuria in kidney transplant recipients. *Nephrol Dial Transplant*. 2011;26(4):1385–1387.
196. Fogazzi GB, Cameron JS, Ritz E, et al. The history of urinary microscopy to the end of the 19th century. *Am J Nephrol*. 1994;14(4–6):452–457.
197. Bottini PV, Lauand JR, Lara Cioffi SG, et al. Glomerular and non-glomerular haematuria: preservation of urine sediment. *Lab Med*. 2005;36(10):3.
198. Kouri TT, Gant VA, Fogazzi GB, et al. Towards European urinalysis guidelines. Introduction of a project under European Confederation of Laboratory Medicine. *Clin Chim Acta*. 2000;297(1–2):305–311.
199. Cohen RA, Brown RS. Clinical practice. Microscopic hematuria. *N Engl J Med*. 2003;348(23):2330–2338.
200. Vivante A, Afek A, Frenkel-Nir Y, et al. Persistent asymptomatic isolated microscopic hematuria in Israeli adolescents and young adults and risk for end-stage renal disease. *JAMA*. 2011;306(7):729–736.
201. Offringa M, Benbassat J. The value of urinary red cell shape in the diagnosis of glomerular and post-glomerular haematuria. A meta-analysis. *Postgrad Med J*. 1992;68(802):648–654.

202. Brodsky SV, Nadasdy T, Rovin BH, et al. Warfarin-related nephropathy occurs in patients with and without chronic kidney disease and is associated with an increased mortality rate. *Kidney Int.* 2011;80(2):181–189.
203. Mishriki SF, Aboumarzouk O, Vint R, et al. Routine urine cytology has no role in hematuria investigations. *J Urol.* 2013;189(4):1255–1258.
204. Wald R, Bell CM, Nisenbaum R, et al. Interobserver reliability of urine sediment interpretation. *Clin J Am Soc Nephrol.* 2009;4(3):567–571.

BOARD REVIEW QUESTIONS

1. A 66-year-old Caucasian male presented to the renal clinic for 6-month follow-up of his known diabetic nephropathy. His baseline creatinine is 180 $\mu\text{mol/L}$ with an estimated glomerular filtration rate (GFR) of 40 mL/min/1.73 m^2 . The trajectory of renal function decline with regard to his estimated GFR (eGFR) has been 2–4 mL/min/1.73 m^2 over the past 4 years. He has had a slowly rising urine albumin-to-creatinine ratio (uACR) and his most recent uACR was 150 mg/mmol . Today, he reports a 2-month history of fatigue, anorexia, and unintentional weight loss of 5–6 kg. Physical examination revealed pulse rate of 80 beats/min and regular blood pressure (140/80 mm Hg), oxygen saturation 95% on room air, clear chest, 1+ pedal edema, and mild pallor.

Laboratory investigations for this clinical visit are as follows:

Hemoglobin (Hb) 103 g/dL , white cell count 8.2 ($\times 10^9/\text{L}$), platelets 230 ($\times 10^9/\text{L}$), creatinine 370 $\mu\text{mol/L}$, urea 18 mmol/L , eGFR 20 mL/min/1.73 m^2 , K 5.7 mmol/L , Na 137 mmol/L , HCO_3^- (bicarbonate) 20 mmol/L , glucose 7.5 mmol/L , and serum albumin 32 g/L . Urinalysis showed 2+ protein and 3+ red blood cells. Urine microscopy revealed a few red cell casts. Which of these is the most appropriate response with regard to this man's condition?

- This is most likely a laboratory error; repeat blood work should be organized.
- He has lost a significant amount of weight in the recent past, which has resulted in a rise in creatinine.
- He has evidence of significant acute kidney injury and needs further investigation.
- This is the natural progression of his underlying diabetic nephropathy.
- All of the above.

Answer: c

Rationale: This patient's history and clinical features suggest acute kidney injury. His subacute onset of symptoms with the recent rise in creatinine, active urine sediment, and presence of red cell casts on urine microscopy strongly suggest the diagnosis of a rapidly progressive glomerulonephritis and he warrants further investigation and management of this condition. Although diabetic nephropathy can rarely progress rapidly, it would be unusual to develop red cell casts in the urine. The weight loss and catabolic state should lead to a decrease in serum creatinine rather than an increase. This result is unlikely to be a laboratory error, especially in the setting of his symptoms, and further investigation and management should not be delayed.

2. A 73-year-old woman was referred to the clinic for evaluation of worsening renal function and anemia. Review of her laboratory tests showed hemoglobin (Hb) 80 g/dL (110 g/dL 1 year ago), creatinine 180 $\mu\text{mol/L}$ (80 $\mu\text{mol/L}$ 1 year ago), urea 15 mmol/L , K 4.6 mmol/L , Na 132 mmol/L , calcium 2.7 mmol/L . Her serum protein electrophoresis showed a monoclonal spike in the gamma region. A diagnosis of multiple myeloma is suspected. An office urine dipstick shows 1+ protein, no red cells, no leukocytes, or glucose. Which of the following statements is true with regard to the urine dipstick result?

- The urine dipstick reliably detects albumin but not other proteins such as monoclonal proteins which cause cast nephropathy.
- The urine dipstick is a sensitive marker for tubular and overflow proteinuria.
- A positive result from a semiquantitative test, such as a urinary dipstick test, should prompt further evaluation with a quantitative test.
- Both a and b.
- Both a and c.

Answer: c

Rationale: This patient has multiple myeloma as evidenced by her history and laboratory findings. Her renal function is impaired, suggesting renal involvement. The urine dipstick shows minimal proteinuria but this test only reliably detects albumin and not other proteins; hence, although it can detect glomerular proteinuria, it is a poor test to assess for tubular and overflow proteinuria. A quantitative test including an albumin-to-creatinine ratio (uACR) and a protein-to-creatinine ratio (uPCR) should be requested. In the case of cast nephropathy, one would expect a significant discrepancy in these values with a higher uPCR as compared with the uACR, suggesting that there is excretion of an alternate protein other than albumin (in this case, light chains).

3. Which of the following statements holds true with regard to estimated glomerular filtration rate (eGFR)?

- The Modification of Diet in Renal Disease (MDRD) study equation is the most commonly used method of determining eGFR and is more accurate than the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation.
- The Cockcroft–Gault (CG) equation remains the most accurate method of determining drug dosing in patients with impaired kidney function.
- The CKD-EPI equation is more accurate than the MDRD study equation and is commonly used to determine eGFR.
- The Berlin Initiative Study group equation can be accurately used to determine estimated glomerular filtration rate (eGFR) for all age groups.

Answer: c

Rationale: The most commonly used methods include the CG equation, the MDRD study equation, and now most recently the CKD-EPI equation. The CKD-EPI equation is more accurate than the MDRD equation, especially at higher glomerular filtration rates (GFRs), and both are more accurate than the CG equation. The latter has not been validated since the adoption of new assay methods for estimation of creatinine and should ideally not be used to determine drug dosing as there is especially a risk of overdosing drugs that have a narrow therapeutic index, with the CG equation being less reliable in assessing the risk of kidney injury. The Berlin Initiative Study equation was used to determine eGFR in the elderly population, and the Full Age Spectrum equation has been used to determine eGFR across all age groups.

4. A 59-year-old man presents to the emergency room with a large-volume upper gastrointestinal (GI) bleed. His

medical history is significant for hypertension and a recent myocardial infarction for which he required percutaneous coronary intervention to the left anterior descending artery. His medications include aspirin (100 mg daily), clopidogrel (75 mg daily), atorvastatin (40 mg daily), ramipril (10 mg daily), and atenolol (25 mg daily). His blood pressure on arrival was 90/60 mm Hg and he was resuscitated with intravenous fluids and packed red blood cells. He is awaiting a gastroscopy. Laboratory investigations showed hemoglobin (Hb) 75 g/dL (baseline 110 g/dL), white cell count $9.8 (\times 10^3/L)$, platelets $200 (\times 10^3/L)$, normal international normalized ratio and partial thromboplastin time, sodium 135 mmol/L, potassium 3.8 mmol/L, urea 30 mmol/L, creatinine 120 $\mu\text{mol/L}$ (baseline creatinine 100 $\mu\text{mol/L}$), and estimated glomerular filtration rate (eGFR) 50 mL/min/1.73 m². Which of the following holds true with regard to his renal function tests?

- a. This patient has acute kidney injury as evidenced by the rise in urea.
- b. This patient has acute kidney injury as evidenced by the decline in eGFR.
- c. Urea is a poor marker of filtration as in this case it is influenced by upper GI bleeding.
- d. Creatinine is an ideal filtration marker as it is freely filtered but not secreted from the renal tubules.
- e. All of the above.

Answer: c

Rationale: Urea is not an accurate filtration marker because it is subject to a number of influences in addition to glomerular filtration. Increased plasma levels may be due to factors independent of renal function and in this case, the elevated urea is due to gastrointestinal bleeding leading to absorption of amino acids from blood in the gastrointestinal tract. Urea and eGFR are not appropriate to diagnose kidney injury in the acute setting. Creatinine is freely filtered by the kidney but is also secreted by the renal tubules and hence, it is not an ideal filtration marker.